Design and Analysis of Experiments
Problem Bank: Detailed Solutions

Solutions to the 40 problems (Chs. 2–11)

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1 Solutions

Solution 1

(a) Write $\sum_j (y_{ij}-\bar y_i)^2 = \sum_j y_{ij}^2 - n_i \bar y_i^2$ (expand the square and use $\sum_j y_{ij} = n_i \bar y_i$). Since $\mathrm{E}(y_{ij}^2) = \mathrm{Var}(y_{ij}) + [\mathrm{E}y_{ij}]^2 = \sigma^2 + \mu_i^2$ and $\mathrm{E}(\bar y_i^2) = \sigma^2/n_i + \mu_i^2$,

$$\mathrm{E}\Big[\sum_j (y_{ij}-\bar y_i)^2\Big] = n_i(\sigma^2+\mu_i^2) - n_i\Big(\frac{\sigma^2}{n_i}+\mu_i^2\Big) = n_i\sigma^2 - \sigma^2 = (n_i-1)\sigma^2.$$

Hence $\mathrm{E}(S_i^2) = \sigma^2$ and

$$\mathrm{E}(S_p^2) = \frac{(n_1-1)\mathrm{E}S_1^2 + (n_2-1)\mathrm{E}S_2^2}{n_1+n_2-2} = \frac{(n_1-1+n_2-1)\sigma^2}{n_1+n_2-2} = \sigma^2.$$

Variance: $S_1^2 \perp\!\!\!\perp S_2^2$, so

$$\mathrm{Var}(S_p^2) = \frac{(n_1-1)^2\frac{2\sigma^4}{n_1-1} + (n_2-1)^2\frac{2\sigma^4}{n_2-1}}{(n_1+n_2-2)^2} = \frac{2\sigma^4(n_1+n_2-2)}{(n_1+n_2-2)^2} = \frac{2\sigma^4}{n_1+n_2-2},$$

which is smaller than $2\sigma^4/(n_i-1)$ because $n_1+n_2-2 > n_i-1$. Pooling uses more information about the common σ^2.

(b) *Numerator.* $\bar y_1 - \bar y_2 \sim N(\mu_1-\mu_2, \sigma^2(\frac{1}{n_1}+\frac{1}{n_2}))$, so under $H_0: \mu_1=\mu_2$

$$Z = \frac{\bar y_1 - \bar y_2}{\sigma\sqrt{1/n_1+1/n_2}} \sim N(0,1).$$

Denominator. $V = \frac{(n_1+n_2-2)S_p^2}{\sigma^2} = \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2}$ is a sum of independent $\chi^2_{n_1-1}$ and $\chi^2_{n_2-1}$, hence $V \sim \chi^2_{n_1+n_2-2}$ (mgf: $(1-2t)^{-(n_1-1)/2}(1-2t)^{-(n_2-1)/2} = (1-2t)^{-(n_1+n_2-2)/2}$). *Independence.* Z is a function of $(\bar y_1, \bar y_2)$, V of (S_1^2, S_2^2); within each sample $\bar y_i \perp\!\!\!\perp S_i^2$ and the samples are independent, so $Z \perp\!\!\!\perp V$. *Conclusion.* By definition of Student's t,

$$t_0 = \frac{Z}{\sqrt{V/(n_1+n_2-2)}} = \frac{\bar y_1-\bar y_2}{\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}\cdot\frac{\sigma}{S_p} = \frac{\bar y_1-\bar y_2}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t_{n_1+n_2-2}.$$

CI. Without H_0, the same argument applied to $(\bar y_1-\bar y_2)-(\mu_1-\mu_2)$ gives a pivot $\sim t_\nu$, $\nu = n_1+n_2-2$:

$$\begin{aligned} 1-\alpha &= P\Big(-t_{\alpha/2,\nu} \le \frac{(\bar y_1-\bar y_2)-(\mu_1-\mu_2)}{S_p\sqrt{1/n_1+1/n_2}} \le t_{\alpha/2,\nu}\Big) \\ &= P\Big((\bar y_1-\bar y_2) - t_{\alpha/2,\nu}S_p\sqrt{\tfrac{1}{n_1}+\tfrac{1}{n_2}} \le \mu_1-\mu_2 \le (\bar y_1-\bar y_2) + t_{\alpha/2,\nu}S_p\sqrt{\tfrac{1}{n_1}+\tfrac{1}{n_2}}\Big). \end{aligned}$$

(c) Let $W = S_1^2/n_1 + S_2^2/n_2$. Then

$$\mathrm{E}W = \frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}, \qquad \mathrm{Var}W = \frac{1}{n_1^2}\cdot\frac{2\sigma_1^4}{n_1-1} + \frac{1}{n_2^2}\cdot\frac{2\sigma_2^4}{n_2-1}.$$

We want $\nu W/\mathrm{E}W \approx \chi^2_\nu$. Its mean is ν automatically; matching variances, $\mathrm{Var}(\nu W/\mathrm{E}W) = \nu^2\mathrm{Var}W/(\mathrm{E}W)^2 \overset{!}{=} 2\nu$, i.e.

$$\nu = \frac{2(\mathrm{E}W)^2}{\mathrm{Var}W} = \frac{\big(\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}\big)^2}{\frac{(\sigma_1^2/n_1)^2}{n_1-1}+\frac{(\sigma_2^2/n_2)^2}{n_2-1}} \xrightarrow{\sigma_i^2\to S_i^2} \hat\nu = \frac{\big(\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}\big)^2}{\frac{(S_1^2/n_1)^2}{n_1-1}+\frac{(S_2^2/n_2)^2}{n_2-1}}.$$

Then $t_0' = Z'/\sqrt{W/\mathrm{E}W}$ with $Z' \sim N(0,1)$ is approximately $t_{\hat\nu}$.

(d) Summary statistics:

$$\sum X = 122.2,\ \bar{X} = 20.367,\ \sum X^2 = 2490.76,$$
$$S_X^2 = \frac{2490.76 - 122.2^2/6}{5} = \frac{2490.76 - 2488.807}{5} = 0.3907,$$
$$\sum Y = 130.6,\ \bar{Y} = 21.767,\ \sum Y^2 = 2844.06,$$
$$S_Y^2 = \frac{2844.06 - 130.6^2/6}{5} = \frac{2844.06 - 2842.727}{5} = 0.2667.$$

$$S_p^2 = \frac{5(0.3907) + 5(0.2667)}{10} = 0.3287, \quad S_p = 0.5733, \quad S_p\sqrt{\tfrac{1}{6} + \tfrac{1}{6}} = 0.5733(0.5774) = 0.3310,$$
$$t_0 = \frac{20.367 - 21.767}{0.3310} = \frac{-1.400}{0.3310} = -4.23.$$

$H_0 : \mu_X = \mu_Y$ vs $H_1 : \mu_X \neq \mu_Y$; null distribution t_{10}; $|t_0| = 4.23 > t_{0.025,10} = 2.228$ ($P = 0.0017$). Reject: the means differ. 95% CI: $-1.400 \pm 2.228(0.3310) = -1.400 \pm 0.737 = [-2.137, -0.663]$ (excludes 0, consistent with the test).

F-test: $F_0 = S_X^2/S_Y^2 = 0.3907/0.2667 = 1.46$; under $H_0 : \sigma_X^2 = \sigma_Y^2$, $F_0 \sim F_{5,5}$; the two-sided 5% region is $F_0 > F_{0.025,5,5} = 7.15$ or $F_0 < 1/7.15 = 0.14$. Do not reject: pooling is justified.

Welch: $S_X^2/6 = 0.06511$, $S_Y^2/6 = 0.04444$, $\hat{\nu} = \frac{(0.10956)^2}{0.06511^2/5 + 0.04444^2/5} = \frac{0.012002}{0.000848 + 0.000395} = 9.66$. Welch t is the same -4.23 here (equal n), compared with $t_{9.66}$: same conclusion.

Solution 2

(a) $d_j = (\mu_1 + \beta_j + \varepsilon_{1j}) - (\mu_2 + \beta_j + \varepsilon_{2j}) = (\mu_1 - \mu_2) + (\varepsilon_{1j} - \varepsilon_{2j})$; β_j cancels. Hence $\mathrm{E}(d_j) = \mu_1 - \mu_2 =: \mu_d$ and $\mathrm{Var}(d_j) = \mathrm{Var}(\varepsilon_{1j}) + \mathrm{Var}(\varepsilon_{2j}) = 2\sigma^2$; the d_j are iid $N(\mu_d, 2\sigma^2)$. By the one-sample theory ($\bar{d} \sim N(\mu_d, 2\sigma^2/n)$, $(n-1)S_d^2/(2\sigma^2) \sim \chi^2_{n-1}$, independent),

$$t_0 = \frac{\bar{d}}{S_d/\sqrt{n}} \overset{H_0:\mu_d=0}{\sim} t_{n-1}.$$

(b) *CRD*: unit u has $\beta_u \sim N(0, \sigma_\beta^2)$, so each observation has variance $\sigma_\beta^2 + \sigma^2$ and all are independent:

$$\mathrm{Var}(\bar{y}_1 - \bar{y}_2)_{\mathrm{CRD}} = \frac{\sigma_\beta^2 + \sigma^2}{n} + \frac{\sigma_\beta^2 + \sigma^2}{n} = \frac{2(\sigma_\beta^2 + \sigma^2)}{n}.$$

Paired: $\mathrm{Var}(\bar{d}) = \frac{2\sigma^2}{n}$. The ratio is $\frac{\sigma_\beta^2 + \sigma^2}{\sigma^2} = 1 + \frac{\sigma_\beta^2}{\sigma^2} \geq 1$, so pairing is never less precise in variance and is much better when specimen-to-specimen variability σ_β^2 is large. The price: the error df drop from $2n-2$ to $n-1$, giving a larger critical value $t_{\alpha/2,n-1}$; pairing loses slightly when $\sigma_\beta^2 \approx 0$.

(c) $d_j = 2, 2, -1, 4, 3, 1$.

$$\bar{d} = \frac{2+2-1+4+3+1}{6} = \frac{11}{6} = 1.833, \qquad \sum d_j^2 = 4+4+1+16+9+1 = 35,$$
$$S_d^2 = \frac{35 - 6(1.833)^2}{5} = \frac{35 - 20.167}{5} = 2.967, \qquad S_d = 1.722,$$
$$t_0 = \frac{1.833}{1.722/\sqrt{6}} = \frac{1.833}{0.7032} = 2.607.$$

$H_0 : \mu_d = 0$ vs $H_1 : \mu_d \neq 0$, $t_0 \sim t_5$ under H_0; $|t_0| = 2.607 > t_{0.025,5} = 2.571$: reject (just) at the 5% level.

(d) Under H_0 the $2^6 = 64$ sign patterns of $(|d_1|, \ldots, |d_6|) = (2, 2, 1, 4, 3, 1)$ are equally likely. $T_{\mathrm{obs}} = 11$ and $\sum |d_j| = 13$. For a pattern with set $\mathcal{N}$ of negative signs, $T = 13 - 2\sum_{j \in \mathcal{N}} |d_j|$. So

$$T \geq 11 \iff \sum_{j \in \mathcal{N}} |d_j| \leq 1 \iff \mathcal{N} = \emptyset \text{ or } \mathcal{N} = \{\text{one of the two 1's}\} : \quad 3 \text{ patterns.}$$

One-sided $p = 3/64 = 0.047$. By symmetry $P(T \leq -11) = 3/64$, so the two-sided $p = 6/64 = 0.094$. The exact test does *not* reject at 5% two-sided, whereas the t-test barely did: with $n = 6$ the normal-theory test relies heavily on normality; the permutation test is exact under symmetry only. (Smallest attainable two-sided p here is $2/64 = 0.031$.)

Solution 3

(a) $y_{ij} - \bar{y}_{\cdot\cdot} = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) + (y_{ij} - \bar{y}_{i\cdot})$. Squaring and summing,

$$SS_T = \sum_i \sum_j (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2 + 2 \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\cdot}).$$

The inner sum is $y_{i\cdot} - n_i \bar{y}_{i\cdot} = 0$ for every i (this is true for any n_i), so the cross term is 0, and $\sum_{j=1}^{n_i} (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = n_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

(b) $S = \sum_i \sum_j (y_{ij} - \mu - \tau_i)^2$; $\partial S/\partial \mu = 0 \Rightarrow N\hat{\mu} + \sum_i n_i \hat{\tau}_i = y_{\cdot\cdot}$; $\partial S/\partial \tau_i = 0 \Rightarrow n_i \hat{\mu} + n_i \hat{\tau}_i = y_{i\cdot}$. The first equation is the sum of the others (rank a). With $\sum n_i \hat{\tau}_i = 0$: $\hat{\mu} = y_{\cdot\cdot}/N = \bar{y}_{\cdot\cdot}$ and $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$. (Check: $\sum n_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) = y_{\cdot\cdot} - N\bar{y}_{\cdot\cdot} = 0$.)

(c) Averages: $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$, and with the weighted constraint $\bar{y}_{\cdot\cdot} = \frac{1}{N} \sum_i n_i (\mu + \tau_i + \bar{\varepsilon}_{i\cdot}) = \mu + \bar{\varepsilon}_{\cdot\cdot}$, where $\bar{\varepsilon}_{\cdot\cdot} = \frac{1}{N} \sum_i n_i \bar{\varepsilon}_{i\cdot}$. So $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})$ and

$$\mathrm{E}(SS_{\mathrm{Trt}}) = \sum_i n_i \tau_i^2 + 2 \sum_i n_i \tau_i \underbrace{\mathrm{E}(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})}_{0} + \mathrm{E} \sum_i n_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2.$$

Using $\sum_i n_i \bar{\varepsilon}_{i\cdot} = N \bar{\varepsilon}_{\cdot\cdot}$,

$$\sum_i n_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2 = \sum_i n_i \bar{\varepsilon}_{i\cdot}^2 - 2\bar{\varepsilon}_{\cdot\cdot} \sum_i n_i \bar{\varepsilon}_{i\cdot} + N\bar{\varepsilon}_{\cdot\cdot}^2 = \sum_i n_i \bar{\varepsilon}_{i\cdot}^2 - N\bar{\varepsilon}_{\cdot\cdot}^2,$$

$$\mathrm{E}[\cdot] = \sum_i n_i \frac{\sigma^2}{n_i} - N\frac{\sigma^2}{N} = a\sigma^2 - \sigma^2 = (a-1)\sigma^2.$$

Therefore $\mathrm{E}(MS_{\mathrm{Trt}}) = \sigma^2 + \frac{\sum n_i \tau_i^2}{a-1}$. For the error, $y_{ij} - \bar{y}_{i\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}$, so by Solution 1(a) $\mathrm{E} \sum_j (y_{ij} - \bar{y}_{i\cdot})^2 = (n_i - 1)\sigma^2$; summing, $\mathrm{E}(SS_E) = (N-a)\sigma^2$ and $\mathrm{E}(MS_E) = \sigma^2$.

(d) If $\sum \tau_i = 0$ but n_i unequal, then $\bar{y}_{\cdot\cdot} = \mu + \bar{\tau}_w + \bar{\varepsilon}_{\cdot\cdot}$ with $\bar{\tau}_w = \sum n_i \tau_i / N \neq 0$ in general, so $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = (\tau_i - \bar{\tau}_w) + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})$ and

$$\mathrm{E}(MS_{\mathrm{Trt}}) = \sigma^2 + \frac{\sum_i n_i (\tau_i - \bar{\tau}_w)^2}{a-1}.$$

Under H_0 (all τ_i equal, hence 0) both versions reduce to σ^2: the test is the same, only the parametrisation of the noncentrality differs.

Solution 4

(a) The effects-coded matrix has columns $\mathbf{1}$ and $\mathbf{x}_k = \mathbf{z}_k - \mathbf{z}_a$ $(k < a)$, where $\mathbf{z}_i$ are the columns of $\mathbf{Z}$; and $\mathbf{1} = \sum_i \mathbf{z}_i$. So every column of $\mathbf{X}$ is a linear combination of columns of $\mathbf{Z}$. Conversely $\mathbf{z}_a = \frac{1}{a}(\mathbf{1} - \sum_{k<a} \mathbf{x}_k)$ (since $\mathbf{1} - \sum_{k<a}(\mathbf{z}_k - \mathbf{z}_a) = a\mathbf{z}_a$) and $\mathbf{z}_k = \mathbf{x}_k + \mathbf{z}_a$. Hence $\Omega(\mathbf{X}) = \Omega(\mathbf{Z})$; both have rank a. The hat matrix is the orthogonal projector onto the column space, which is unique, so $\mathbf{H_X} = \mathbf{H_Z}$.

(b) $\mathbf{Z}^T\mathbf{Z} = (\mathbf{I}_a \otimes \mathbf{1}_n^T)(\mathbf{I}_a \otimes \mathbf{1}_n) = \mathbf{I}_a \otimes (\mathbf{1}_n^T\mathbf{1}_n) = n\mathbf{I}_a$. Hence

$$\mathbf{H} = \mathbf{Z}\frac{1}{n}\mathbf{I}_a\mathbf{Z}^T = \frac{1}{n}(\mathbf{I}_a \otimes \mathbf{1}_n)(\mathbf{I}_a \otimes \mathbf{1}_n^T) = \frac{1}{n}\mathbf{I}_a \otimes \mathbf{1}_n\mathbf{1}_n^T = \frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n.$$

$\mathbf{Hy}$ replaces every observation by its treatment mean: $(\mathbf{Hy})_{ij} = \frac{1}{n}\sum_{j'} y_{ij'} = \bar{y}_{i\cdot}$.

(c) $\mathbf{H}^T = \mathbf{H}$ clearly; $\mathbf{HH} = \frac{1}{n^2}\mathbf{I}_a \otimes \mathbf{J}_n\mathbf{J}_n = \frac{1}{n^2}\mathbf{I}_a \otimes n\mathbf{J}_n = \mathbf{H}$. With $\mathbf{J}_N = \mathbf{J}_a \otimes \mathbf{J}_n$:

$$\mathbf{H} \cdot \tfrac{1}{N}\mathbf{J}_N = \tfrac{1}{nN}\mathbf{J}_a \otimes \mathbf{J}_n\mathbf{J}_n = \tfrac{1}{N}\mathbf{J}_a \otimes \mathbf{J}_n = \tfrac{1}{N}\mathbf{J}_N,$$

and similarly $\frac{1}{N}\mathbf{J}_N\mathbf{H} = \frac{1}{N}\mathbf{J}_N$, $\frac{1}{N^2}\mathbf{J}_N^2 = \frac{1}{N}\mathbf{J}_N$, so $(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)^2 = \mathbf{H} - \frac{2}{N}\mathbf{J}_N + \frac{1}{N}\mathbf{J}_N = \mathbf{H} - \frac{1}{N}\mathbf{J}_N$. Traces: $\mathrm{tr}\mathbf{H} = \frac{1}{n} \cdot a \cdot n = a$; $\mathrm{tr}(\mathbf{H} - \frac{1}{N}\mathbf{J}_N) = a - 1$; $\mathrm{tr}(\mathbf{I} - \mathbf{H}) = N - a$.

(d) $(\mathbf{I}-\mathbf{H})(\mathbf{H}-\frac{1}{N}\mathbf{J}_N) = \mathbf{H}-\frac{1}{N}\mathbf{J}_N-\mathbf{H}+\frac{1}{N}\mathbf{J}_N = \mathbf{0}$. Write $\mathbf{y}-\bar{y}_{\cdot\cdot}\mathbf{1} = (\mathbf{I}-\frac{1}{N}\mathbf{J}_N)\mathbf{y} = (\mathbf{H}-\frac{1}{N}\mathbf{J}_N)\mathbf{y}+(\mathbf{I}-\mathbf{H})\mathbf{y}$. The two pieces are orthogonal vectors (their inner product is $\mathbf{y}^T(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)(\mathbf{I}-\mathbf{H})\mathbf{y} = 0$), so by Pythagoras

$$\|\mathbf{y} - \bar{y}_{\cdot\cdot}\mathbf{1}\|^2 = \|(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N)\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{H})\mathbf{y}\|^2, \quad \text{i.e. } SS_T = SS_{\mathrm{Trt}} + SS_E.$$

Solution 5

(a) $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$. By $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \mathrm{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$ with $\mathbf{A} = (\mathbf{I} - \mathbf{H})^T(\mathbf{I} - \mathbf{H}) = \mathbf{I} - \mathbf{H}$, $\mathbf{V} = \sigma^2\mathbf{I}$, $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$:

$$\mathrm{E}(\mathbf{e}^T\mathbf{e}) = \sigma^2\mathrm{tr}(\mathbf{I} - \mathbf{H}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \sigma^2(N - a) + 0,$$

since $\mathrm{tr}\mathbf{H} = \mathrm{tr}\big((\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\big) = \mathrm{tr}\mathbf{I}_a = a$ and $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$.

$$\mathrm{Cov}(\hat{\mathbf{y}}, \mathbf{e}) = \mathrm{Cov}(\mathbf{H}\mathbf{y}, (\mathbf{I} - \mathbf{H})\mathbf{y}) = \mathbf{H}(\sigma^2\mathbf{I})(\mathbf{I} - \mathbf{H})^T = \sigma^2(\mathbf{H} - \mathbf{H}^2) = \mathbf{0}.$$

(b) $\mathrm{Var}(\hat{\mathbf{y}}) = \mathbf{H}\sigma^2\mathbf{I}\mathbf{H}^T = \sigma^2\mathbf{H}$, so $\sum_\ell \mathrm{Var}(\hat{y}_\ell) = \sigma^2\mathrm{tr}(\mathbf{H}) = a\sigma^2$: $\kappa = a$ (the number of parameters). Directly: $\hat{y}_{ij} = \bar{y}_{i\cdot}$ has variance σ^2/n; there are $N = an$ fitted values, so the sum is $an \cdot \sigma^2/n = a\sigma^2$.

(c) (i) $SS_{\mathrm{Trt}} = \mathbf{y}^T\mathbf{A}\mathbf{y}$, $SS_E = \mathbf{y}^T\mathbf{B}\mathbf{y}$ with $\mathbf{A} = (\mathbf{H} - \frac{1}{N}\mathbf{J})/\sigma^2$, $\mathbf{B} = (\mathbf{I} - \mathbf{H})/\sigma^2$ (both σ^2-scaled idempotent, so both forms are χ^2 by Theorem 3.11), and $\mathbf{A}(\sigma^2\mathbf{I})\mathbf{B} = \frac{1}{\sigma^2}(\mathbf{H} - \frac{1}{N}\mathbf{J})(\mathbf{I} - \mathbf{H}) = \mathbf{0}$ (Solution 4(d)); Theorem 3.12 gives independence.

(ii) $SS_{\mathrm{Trt}} = n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$ is a function of $(\bar{y}_{1\cdot}, \ldots, \bar{y}_{a\cdot})$ only ($\bar{y}_{\cdot\cdot}$ is their average). $SS_E = \sum_i(n - 1)S_i^2$ is a function of $(S_1^2, \ldots, S_a^2)$. For a normal sample $\bar{y}_{i\cdot} \perp\!\!\!\perp S_i^2$, and different treatments are independent, so the vector $(\bar{y}_{1\cdot}, \ldots, \bar{y}_{a\cdot})$ is independent of $(S_1^2, \ldots, S_a^2)$; functions of independent vectors are independent.

(d) $SS_E/\sigma^2 \sim \chi^2_{N-a}$; $SS_{\mathrm{Trt}}/\sigma^2 \sim \chi^2_{a-1,\lambda}$ with $\lambda = \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H} - \frac{1}{N}\mathbf{J})\mathbf{X}\boldsymbol{\beta}/\sigma^2 = \|(\mathbf{H} - \frac{1}{N}\mathbf{J})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$. Since $(\mathbf{H} - \frac{1}{N}\mathbf{J})\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta} - \mu\mathbf{1}$ has entries τ_i (each repeated n times), $\lambda = n\sum\tau_i^2/\sigma^2$. The ratio of independent χ^2's over their df gives $F_0 \sim F_{a-1,N-a,\lambda}$; under H_0, $\lambda = 0$.

Solution 6

(a) $\mathrm{E}(Z_i^2) = \mathrm{Var}(Z_i) + (\mathrm{E}Z_i)^2 = 1 + \delta_i^2$, so $\mathrm{E}(U) = \sum_{i=1}^{k}(1+\delta_i^2) = k + \lambda$.

(b) With density $f(v) = \frac{v^{m/2-1}e^{-v/2}}{2^{m/2}\Gamma(m/2)}$,

$$\mathrm{E}(V^{-1}) = \int_0^\infty \frac{v^{m/2-2}e^{-v/2}}{2^{m/2}\Gamma(m/2)}dv = \frac{2^{m/2-1}\Gamma(m/2-1)}{2^{m/2}\Gamma(m/2)} = \frac{\Gamma(m/2-1)}{2\,(m/2-1)\Gamma(m/2-1)} = \frac{1}{m-2}.$$

(c) $F_0 = \frac{U/(a-1)}{V/(N-a)}$ with $U = SS_{\mathrm{Trt}}/\sigma^2 \sim \chi^2_{a-1,\lambda}$, $V = SS_E/\sigma^2 \sim \chi^2_{N-a}$, independent. Hence

$$\mathrm{E}(F_0) = \frac{N-a}{a-1}\mathrm{E}(U)\mathrm{E}(V^{-1}) = \frac{N-a}{a-1}\cdot(a-1+\lambda)\cdot\frac{1}{N-a-2} = \frac{N-a}{N-a-2}\Big(1+\frac{\lambda}{a-1}\Big).$$

Under H_0, $\mathrm{E}(F_0) = \frac{N-a}{N-a-2} > 1$, whereas $\mathrm{E}(MS_{\mathrm{Trt}})/\mathrm{E}(MS_E) = 1$: the expectation of a ratio is not the ratio of expectations ($\mathrm{E}(1/V) > 1/\mathrm{E}V$ by Jensen). Still, $\frac{N-a}{N-a-2} \to 1$ as the error df grow, and the ratio of EMS is what tells us the *direction*: F_0 is centred near 1 under H_0 and pushed upward by λ under H_1.

(d) $\sum \tau_i^2 = 4$, $\lambda = \frac{5\cdot 4}{4} = 5$; $N - a = 16$: $\mathrm{E}(F_0) = \frac{16}{14}(1+\frac{5}{3}) = 1.1429 \times 2.6667 = 3.05$. In general $\lambda = n \cdot 4/4 = n$, so $\lambda \geq 10 \iff n \geq 10$.

Solution 7

(a) $C = \sum c_i y_{i\cdot}$ is normal with $\mathrm{E}(C) = n\sum c_i\mu_i$ and $\mathrm{Var}(C) = \sum c_i^2 \mathrm{Var}(y_{i\cdot}) = n\sigma^2\sum c_i^2$. Under $H_0 : \sum c_i\mu_i = 0$, $C/\sqrt{n\sigma^2\sum c_i^2} \sim N(0,1)$, hence $\frac{C^2}{n\sigma^2\sum c_i^2} = \frac{SS_C}{\sigma^2} \sim \chi^2_1$ (one df: a single linear function). (Note $\sum c_i\mu_i = \sum c_i\tau_i$ because $\sum c_i = 0$.)

(b) $\mathrm{Cov}(C,D) = \sum_i\sum_{i'} c_i d_{i'}\mathrm{Cov}(y_{i\cdot}, y_{i'\cdot}) = \sum_i c_i d_i\, n\sigma^2 = 0$. (C,D) is bivariate normal, so zero covariance $\Rightarrow$ independence.

(c) Let $\mathbf{z} = \sqrt{n}(\bar y_{1\cdot},\ldots,\bar y_{a\cdot})^T$, so $SS_{\mathrm{Trt}} = n\sum(\bar y_{i\cdot} - \bar y_{\cdot\cdot})^2 = \|(\mathbf{I}_a - \frac{1}{a}\mathbf{J}_a)\mathbf{z}\|^2$. Let $\mathbf{q}_0 = \mathbf{1}/\sqrt{a}$ and $\mathbf{q}_k = \mathbf{c}_k/\|\mathbf{c}_k\|$; these a vectors are orthonormal ($\mathbf{c}_k \perp \mathbf{1}$ because $\sum c_{ki} = 0$, and $\mathbf{c}_k \perp \mathbf{c}_l$). Then $\mathbf{I}_a - \frac{1}{a}\mathbf{J}_a = \sum_{k=1}^{a-1}\mathbf{q}_k\mathbf{q}_k^T$ (projection onto $\mathbf{1}^\perp$), so

$$SS_{\mathrm{Trt}} = \sum_{k=1}^{a-1}(\mathbf{q}_k^T\mathbf{z})^2 = \sum_{k=1}^{a-1}\frac{\left(\sqrt{n}\sum_i c_{ki}\bar y_{i\cdot}\right)^2}{\sum_i c_{ki}^2} = \sum_{k=1}^{a-1}\frac{(\sum_i c_{ki}y_{i\cdot})^2}{n\sum_i c_{ki}^2} = \sum_k SS_{C_k}.$$

(d) Totals $(49, 77, 88, 108, 54)$, $n = 5$.

$$C_L = -2(49) - 77 + 0 + 108 + 2(54) = 41, \qquad SS_L = \tfrac{41^2}{5\cdot 10} = 33.62,$$
$$C_Q = 2(49) - 77 - 2(88) - 108 + 2(54) = -155, \qquad SS_Q = \tfrac{155^2}{5\cdot 14} = 343.21,$$
$$C_C = -49 + 2(77) + 0 - 2(108) + 54 = -57, \qquad SS_C = \tfrac{57^2}{5\cdot 10} = 64.98,$$
$$C_4 = 49 - 4(77) + 6(88) - 4(108) + 54 = -109, \qquad SS_4 = \tfrac{109^2}{5\cdot 70} = 33.95.$$

Sum: $33.62 + 343.21 + 64.98 + 33.95 = 475.76 = SS_{\mathrm{Trt}}$ ✓. Tests ($F_{0.05,1,20} = 4.35$, $MS_E = 8.06$): linear 4.17 (n.s.), quadratic 42.58 (*), cubic 8.06 (*), quartic 4.21 (n.s.). The response is strongly curved (maximum near 30%), with a significant cubic component.

Solution 8

(a) Let $u_i = \bar{y}_{i\cdot} - \mu_i \overset{\text{iid}}{\sim} N(0, \sigma^2/n)$ and $\bar{u} = \frac{1}{a}\sum u_i = \bar{y}_{\cdot\cdot} - \bar{\mu}$. Because $\sum c_i = 0$,

$$C_{\mathbf{c}} - \Gamma_{\mathbf{c}} = \sum_i c_i u_i = \sum_i c_i (u_i - \bar{u}).$$

Cauchy–Schwarz: $\big(\sum c_i(u_i - \bar{u})\big)^2 \le \sum c_i^2 \sum (u_i - \bar{u})^2$, with equality iff $c_i \propto u_i - \bar{u}$ (an admissible contrast, as $\sum(u_i - \bar{u}) = 0$). Dividing by $\frac{\sigma^2}{n}\sum c_i^2$,

$$\frac{(C_{\mathbf{c}} - \Gamma_{\mathbf{c}})^2}{\frac{\sigma^2}{n}\sum c_i^2} \le \frac{n\sum_i (u_i - \bar{u})^2}{\sigma^2}.$$

(b) $\sqrt{n}u_i/\sigma \overset{\text{iid}}{\sim} N(0,1)$, so $\frac{n\sum(u_i - \bar{u})^2}{\sigma^2} \sim \chi^2_{a-1}$ (sample-variance result). It depends only on the treatment means, so it is independent of SS_E (Solution 5(c)(ii)), and $(N-a)MS_E/\sigma^2 \sim \chi^2_{N-a}$. Hence $\frac{n\sum(u_i - \bar{u})^2/(a-1)}{MS_E} \sim F_{a-1,N-a}$ for every $\boldsymbol{\mu}$ (the μ_i are subtracted).

(c) Taking the supremum over $\mathbf{c}$ in (a) (equality is attained) and replacing σ^2 by MS_E:

$$\sup_{\mathbf{c}\in\mathcal{C}} \frac{(C_{\mathbf{c}} - \Gamma_{\mathbf{c}})^2}{MS_E \sum c_i^2/n} = (a-1)\mathcal{F}, \quad \mathcal{F} \sim F_{a-1,N-a}.$$

So $P\big(|C_{\mathbf{c}} - \Gamma_{\mathbf{c}}| \le \sqrt{(a-1)F_\alpha}\sqrt{MS_E \sum c_i^2/n}\ \forall \mathbf{c}\big) = P\big((a-1)\mathcal{F} \le (a-1)F_\alpha\big) = 1 - \alpha$.

(d) Under H_0 all $\Gamma_{\mathbf{c}} = 0$ and $\sup_{\mathbf{c}} \frac{C_{\mathbf{c}}^2}{MS_E \sum c_i^2/n} = \frac{n\sum(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}{MS_E} = (a-1)F_0$ (attained at $c_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$). Some contrast is declared significant $\iff$ sup $> (a-1)F_\alpha \iff F_0 > F_\alpha$. The Scheffé procedure is exactly as powerful as (and consistent with) the overall F-test, which explains its conservativeness for individual contrasts.

Solution 9

(a) Induction: $(1-\alpha)^1 = 1-\alpha$; if $(1-\alpha)^r \geq 1-r\alpha$, then $(1-\alpha)^{r+1} \geq (1-r\alpha)(1-\alpha) = 1-(r+1)\alpha + r\alpha^2 \geq 1-(r+1)\alpha$. Without independence, by Boole's inequality $P(\cup_i E_i) \leq \sum_i P(E_i) = r\alpha$. For $r = 6$: $\alpha_{FW} = 1 - 0.95^6 = 1 - 0.7351 = 0.265$ (≤ 0.30).

(b) Totals $100, 125, 110, 140$ (means $20, 25, 22, 28$), $y_{\cdot\cdot} = 475$, $\sum y^2 = 11513$, $CF = 475^2/20 = 11281.25$.

$$SS_T = 11513 - 11281.25 = 231.75,$$
$$SS_{\text{Trt}} = \frac{100^2 + 125^2 + 110^2 + 140^2}{5} - 11281.25 = \frac{57325}{5} - 11281.25 = 11465 - 11281.25 = 183.75,$$
$$SS_E = 231.75 - 183.75 = 48.00.$$

Source	SS	DF	MS	F_0
Treatments	183.75	3	61.25	20.42
Error	48.00	16	3.00	
Total	231.75	19		

$F_0 = 20.42 > F_{0.05,3,16} = 3.24$: reject $H_0 : \mu_1 = \cdots = \mu_4$.

(c) $\sqrt{2MS_E/n} = \sqrt{1.2} = 1.095$, $\sqrt{MS_E/n} = 0.7746$.

- LSD: $t_{0.025,16}(1.095) = 2.120(1.095) = 2.32$.
- Bonferroni: $t_{0.05/12,16} = 3.008$, critical difference $3.008(1.095) = 3.30$.
- Tukey: $T = 4.05(0.7746) = 3.14$.

Pair	$\lvert\bar{y}_i - \bar{y}_j\rvert$	LSD (2.32)	Tukey (3.14)	Bonf. (3.30)
4–1	8	*	*	*
4–3	6	*	*	*
2–1	5	*	*	*
4–2	3	*	–	–
2–3	3	*	–	–
3–1	2	–	–	–

Groupings (means in increasing order $\bar{y}_1 = 20 < \bar{y}_3 = 22 < \bar{y}_2 = 25 < \bar{y}_4 = 28$): LSD: 1 3 2 4 (only 1 and 3 not separated); Tukey and Bonferroni: 1 3, 3 2, 2 4 (overlapping groups: 1–2, 1–4 and 3–4 are separated, the other three pairs are not). Bonferroni is the most conservative here ($3.30 > 3.14 > 2.32$), Tukey is exact for all pairwise comparisons and therefore shorter than Bonferroni.

(d) $C = 28 - \frac{1}{3}(20+25+22) = 28 - 22.333 = 5.667$, $\sum c_i^2 = 1 + 3 \cdot \frac{1}{9} = \frac{4}{3}$, $se = \sqrt{3 \cdot \frac{4}{3}/5} = \sqrt{0.8} = 0.894$. t-test: $t_0 = 5.667/0.894 = 6.34 > t_{0.025,16} = 2.120$. Scheffé: $S = \sqrt{3F_{0.05,3,16}}\,(0.894) = \sqrt{3(3.24)}(0.894) = 3.117(0.894) = 2.79 < 5.667$: significant too. If the contrast was chosen *after* seeing that $\bar{y}_4$ is largest (data snooping), only Scheffé protects the error rate, because it is valid simultaneously for all contrasts.

Solution 10

(a) Ranks are $1,\dots,N$: $\bar R = \frac{1}{N}\frac{N(N+1)}{2} = \frac{N+1}{2}$, and

$$\sum_{k=1}^{N}(k-\bar R)^2 = \sum k^2 - N\bar R^2 = \frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)^2}{4}$$
$$= \frac{N(N+1)[2(2N+1)-3(N+1)]}{12} = \frac{N(N+1)(N-1)}{12},$$

so $S^2 = N(N+1)/12$. Next, with $R_{i\cdot} = n_i\bar R_{i\cdot}$ and $\sum R_{i\cdot} = N(N+1)/2$,

$$\sum_i n_i\Big(\bar R_{i\cdot} - \frac{N+1}{2}\Big)^2 = \sum_i \frac{R_{i\cdot}^2}{n_i} - (N+1)\sum_i R_{i\cdot} + N\frac{(N+1)^2}{4} = \sum_i \frac{R_{i\cdot}^2}{n_i} - \frac{N(N+1)^2}{4}.$$

Dividing by $S^2 = N(N+1)/12$ gives the result: H is the treatment SS of the ranks divided by the (known) variance of a rank.

(b) Pooled ordering: $2.8(1), 3.1(2), 4.5(3), 4.9(4), 5.6(5), 5.9(6), 6.1(7), 7.2(8), 8.0(9)$. Rank sums: G1: $2+3+1=6$; G2: $5+7+4=16$; G3: $8+6+9=23$ (total 45 ✓).

$$H = \frac{12}{9\cdot 10}\Big[\frac{6^2+16^2+23^2}{3}\Big] - 3(10) = \frac{12}{90}\cdot\frac{821}{3} - 30 = 36.489 - 30 = 6.49.$$

($\frac{12}{N(N+1)}\sum\frac{R_{i\cdot}^2}{n_i} - 3(N+1)$ is the same formula expanded.) $H = 6.49 > \chi^2_{0.05,2} = 5.99$: reject equality of the three distributions ($P \approx 0.039$; with $n_i = 3$ the χ^2 approximation is rough).

(c) Expanding $h(y) = h(\mu) + h'(\mu)(y-\mu) + \cdots$, $\mathrm{Var}h(y) \approx h'(\mu)^2 w(\mu)$. Setting this constant $= K$: $h'(\mu) = \sqrt{K}\, w(\mu)^{-1/2}$, $h(\mu) = \sqrt{K}\int w(\mu)^{-1/2}d\mu$. (i) Poisson, $w = \mu$: $h \propto \int \mu^{-1/2} = 2\sqrt{\mu} \Rightarrow \sqrt{y}$. (ii) $\sigma \propto \mu$, $w \propto \mu^2$: $h \propto \int \mu^{-1} = \log\mu \Rightarrow \log y$. (iii) $\sigma \propto \mu^2$, $w \propto \mu^4$: $h \propto \int \mu^{-2} = -\mu^{-1} \Rightarrow 1/y$. (iv) $\sigma \propto \mu^\alpha$, $w \propto \mu^{2\alpha}$: $h \propto \int \mu^{-\alpha} = \frac{\mu^{1-\alpha}}{1-\alpha} \Rightarrow y^{1-\alpha}$ ($\alpha \neq 1$), $\log y$ ($\alpha = 1$).

(d) $h(y) = \sqrt{y}$, $h'(\mu) = \frac{1}{2\sqrt{\mu}}$: $\mathrm{Var}(\sqrt{y}) \approx \frac{1}{4\mu}\cdot\mu = \frac{1}{4}$, independent of μ.

Solution 11

(a) $S = \sum_i \sum_j (y_{ij} - \mu - \tau_i)^2$ gives

$$N\hat{\mu} + n\sum_i \hat{\tau}_i = y_{\cdot\cdot}, \qquad n\hat{\mu} + n\hat{\tau}_i = y_{i\cdot}\ (i = 1, \ldots, a).$$

Summing the last a equations reproduces the first, so only a equations are independent for $a+1$ unknowns: one constraint is needed.

- $\tau_a = 0$: the ath equation gives $n\hat{\mu} = y_{a\cdot}$, so $\hat{\mu} = \bar{y}_{a\cdot}$ and then $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{a\cdot}$.
- $\sum \tau_i = 0$: the first equation gives $N\hat{\mu} = y_{\cdot\cdot}$, so $\hat{\mu} = \bar{y}_{\cdot\cdot}$ and $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$.

(b) Under both constraints

$$\hat{\mu} + \hat{\tau}_i = \bar{y}_{i\cdot}, \qquad \hat{\tau}_i - \hat{\tau}_j = \bar{y}_{i\cdot} - \bar{y}_{j\cdot}, \qquad \sum_i c_i\hat{\tau}_i = \sum_i c_i\bar{y}_{i\cdot} - (\text{const})\underbrace{\sum_i c_i}_{=0} = \sum_i c_i\bar{y}_{i\cdot}.$$

These coincide, whereas $\hat{\mu}$ ($\bar{y}_{a\cdot}$ vs $\bar{y}_{\cdot\cdot}$) and $\hat{\tau}_i$ ($\bar{y}_{i\cdot} - \bar{y}_{a\cdot}$ vs $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$) depend on the arbitrary constraint. A linear function $\boldsymbol{\lambda}^T\boldsymbol{\beta}$ is *estimable* if it equals $\mathrm{E}(\mathbf{t}^T\mathbf{y})$ for some $\mathbf{t}$; here $\mathrm{E}(\bar{y}_{i\cdot}) = \mu + \tau_i$ and $\mathrm{E}(\sum c_i\bar{y}_{i\cdot}) = \sum c_i\tau_i$, so these are estimable, and the LSE of an estimable function is unique. μ alone is not estimable: $\mathrm{E}(\mathbf{t}^T\mathbf{y}) = \mu\sum t_{ij} + \sum_i \tau_i \sum_j t_{ij}$ can never isolate μ for all τ.

(c) Rows (levels 1,1,2,2,3,3,4,4):

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & -1 \end{pmatrix}, \quad \mathbf{X}^T\mathbf{X} = \begin{pmatrix} 8 & 0 & 0 & 0 \\ 0 & 4 & 2 & 2 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 4 \end{pmatrix}, \quad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} 0 \\ y_{1\cdot} - y_{4\cdot} \\ y_{2\cdot} - y_{4\cdot} \\ y_{3\cdot} - y_{4\cdot} \end{pmatrix} = \begin{pmatrix} 0 \\ 20 \\ 10 \\ 10 \end{pmatrix},$$

since $y_{1\cdot} = 10$, $y_{2\cdot} = y_{3\cdot} = 0$, $y_{4\cdot} = -10$. The lower block is $2(\mathbf{I}_3 + \mathbf{J}_3)$ with inverse $\frac{1}{2}(\mathbf{I}_3 - \frac{1}{4}\mathbf{J}_3)$ (check: $(\mathbf{I}+\mathbf{J})(\mathbf{I} - \frac{1}{4}\mathbf{J}) = \mathbf{I} + \mathbf{J} - \frac{1}{4}\mathbf{J} - \frac{3}{4}\mathbf{J} = \mathbf{I}$). Hence

$$\hat{\mu} = \frac{0}{8} = 0, \qquad \begin{pmatrix} \hat{\tau}_1 \\ \hat{\tau}_2 \\ \hat{\tau}_3 \end{pmatrix} = \frac{1}{2}\Big[\begin{pmatrix} 20 \\ 10 \\ 10 \end{pmatrix} - \frac{40}{4}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\Big] = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix}, \qquad \hat{\tau}_4 = -(5+0+0) = -5.$$

Fitted values $\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i$: $\hat{\mathbf{y}} = (5, 5, 0, 0, 0, 0, -5, -5)^T = \mathbf{y}$, so $\mathbf{e} = \mathbf{0}$ and $SS_E = 0$: every observation equals its treatment mean (no within-treatment variation). Check: $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ with $\bar{y}_{\cdot\cdot} = 0$ and $\bar{y}_{i\cdot} = (5, 0, 0, -5)$.

(d) Dummy columns $\mathbb{1}$(level k), $k = 2, 3, 4$, together with $\mathbf{1}$ span the same space as the indicator columns $\mathbf{z}_1, \ldots, \mathbf{z}_4$, as do the effects-coded columns (Solution 4(a)). The hat matrix is the orthogonal projection onto that common column space, hence identical; so are $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $SS_E = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$. Only the meaning of the coefficients changes ($\mathbf{X}_2 = \mathbf{X}_1\mathbf{T}$ for an invertible $\mathbf{T}$, so $\hat{\boldsymbol{\beta}}_2 = \mathbf{T}^{-1}\hat{\boldsymbol{\beta}}_1$).

Solution 12

(a) Write $y_{ij} - \bar{y}_{\cdot\cdot} = A_i + B_j + C_{ij}$, $A_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $B_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $C_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. The cross terms are

$$\sum\sum A_i B_j = \Big(\sum_i A_i\Big)\Big(\sum_j B_j\Big) = 0, \qquad \sum_j B_j \sum_i C_{ij} = 0,$$
$$\sum_i A_i \sum_j C_{ij} = \sum_i A_i(b\bar{y}_{i\cdot} - b\bar{y}_{i\cdot} - b\bar{y}_{\cdot\cdot} + b\bar{y}_{\cdot\cdot}) = 0.$$

Hence $SS_T = b\sum A_i^2 + a\sum B_j^2 + \sum\sum C_{ij}^2$. The ab residuals satisfy a row and b column zero-sum restrictions, of which $a + b - 1$ are independent (the grand total is counted twice), so df $= ab - a - b + 1 = (a-1)(b-1)$.

(b) $\bar{y}_{i\cdot} = \mu + \tau_i + \frac{1}{b}\sum_j \beta_j + \bar{\varepsilon}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$; $\bar{y}_{\cdot\cdot} = \mu + \bar{\varepsilon}_{\cdot\cdot}$. So $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})$ and

$$\begin{aligned}\mathrm{E}(SS_{\mathrm{Trt}}) &= b\sum_i \tau_i^2 + 2b\sum_i \tau_i \mathrm{E}(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}) + b\,\mathrm{E}\sum_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2 \\ &= b\sum_i \tau_i^2 + 0 + b(a-1)\frac{\sigma^2}{b},\end{aligned}$$

because $\bar{\varepsilon}_{1\cdot}, \ldots, \bar{\varepsilon}_{a\cdot}$ are iid with variance σ^2/b and $\mathrm{E}\sum_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2 = (a-1)\sigma^2/b$. Dividing by $a-1$: $\mathrm{E}(MS_{\mathrm{Trt}}) = \sigma^2 + \frac{b}{a-1}\sum \tau_i^2$.

(c) Substituting the model, $e_{ij} = (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}) - (\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})$. Then $SS_E = ① + ② - 2\,③$ with

$$\mathrm{E}① = \mathrm{E}\sum_i\sum_j (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2 = \sum_{i=1}^{a}(b-1)\sigma^2 = a(b-1)\sigma^2,$$
$$\mathrm{E}② = \mathrm{E}\sum_i\sum_j (\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2 = a\cdot(b-1)\frac{\sigma^2}{a} = (b-1)\sigma^2,$$
$$③ = \sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}) = \underbrace{\sum\sum \varepsilon_{ij}\bar{\varepsilon}_{\cdot j}}_{(i)} - \underbrace{\sum\sum \varepsilon_{ij}\bar{\varepsilon}_{\cdot\cdot}}_{(ii)} - \underbrace{\sum\sum \bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j}}_{(iii)} + \underbrace{\sum\sum \bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot\cdot}}_{(iv)}.$$

- (i) $= \sum_j \bar{\varepsilon}_{\cdot j}\sum_i \varepsilon_{ij} = a\sum_j \bar{\varepsilon}_{\cdot j}^2$, $\mathrm{E} = a\cdot b\cdot\frac{\sigma^2}{a} = b\sigma^2$.
- (ii) $= ab\,\bar{\varepsilon}_{\cdot\cdot}^2$, $\mathrm{E} = ab\cdot\frac{\sigma^2}{ab} = \sigma^2$; (iv) $= \bar{\varepsilon}_{\cdot\cdot}\, b\sum_i \bar{\varepsilon}_{i\cdot} = ab\,\bar{\varepsilon}_{\cdot\cdot}^2$, $\mathrm{E} = \sigma^2$.
- (iii) $\mathrm{E}(\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j}) = \mathrm{E}\big[\frac{1}{b}\sum_{j'}\varepsilon_{ij'}\cdot\frac{1}{a}\sum_{i'}\varepsilon_{i'j}\big] = \frac{1}{ab}\mathrm{E}(\varepsilon_{ij}^2) = \frac{\sigma^2}{ab}$ (only the common term ε_{ij} survives); summing over the ab pairs, $\mathrm{E}(iii) = \sigma^2$.

So $\mathrm{E}③ = b\sigma^2 - \sigma^2 - \sigma^2 + \sigma^2 = (b-1)\sigma^2$ and

$$\mathrm{E}(SS_E) = a(b-1)\sigma^2 + (b-1)\sigma^2 - 2(b-1)\sigma^2 = (a-1)(b-1)\sigma^2.$$

(d) $SS_E = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$ with $\mathrm{rank}(\mathbf{X}) = 1 + (a-1) + (b-1) = a + b - 1$, $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, $\mathbf{V} = \sigma^2\mathbf{I}$: $\mathrm{E}(SS_E) = \sigma^2\mathrm{tr}(\mathbf{I} - \mathbf{H}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \sigma^2(ab - a - b + 1) + 0$. ✓

(e) Normal equations: $ab\hat{\mu} + b\sum\hat{\tau}_i + a\sum\hat{\beta}_j = y_{\cdot\cdot}$; $b\hat{\mu} + b\hat{\tau}_i + \sum_j \hat{\beta}_j = y_{i\cdot}$; $a\hat{\mu} + \sum_i \hat{\tau}_i + a\hat{\beta}_j = y_{\cdot j}$. With the zero-sum constraints: $\hat{\mu} = \bar{y}_{\cdot\cdot}$, $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $\hat{\beta}_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$. Then $\hat{\tau}_i - \hat{\tau}_k = \bar{y}_{i\cdot} - \bar{y}_{k\cdot}$; the two means use disjoint observations, each with variance σ^2/b (the β_j are constants), so $\mathrm{Var}(\hat{\tau}_i - \hat{\tau}_k) = 2\sigma^2/b$.

Solution 13

(a) The fitted values are $\hat{y}_{ij}^{(F)} = \bar{y}_{i\cdot} + \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$ and $\hat{y}_{ij}^{(R)} = \bar{y}_{\cdot j}$ (reduced model = one-way model in blocks), so

$$SS_{E(F)} = \sum\sum (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2, \qquad SS_{E(R)} = \sum\sum (y_{ij} - \bar{y}_{\cdot j})^2.$$

Write $y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{\cdot j}) - (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})$:

$$\begin{aligned} SS_{E(F)} &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \sum_j (y_{ij} - \bar{y}_{\cdot j}) \\ &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \cdot b(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \\ &= SS_{E(R)} - b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2, \end{aligned}$$

using $\sum_j (y_{ij} - \bar{y}_{\cdot j}) = y_{i\cdot} - y_{\cdot\cdot}/a = b(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})$. Hence $SS_H = b\sum(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_{\text{Trt}}$.

(b) Every column of $\mathbf{X}_R$ is a column of $\mathbf{X}$, so $\Omega(\mathbf{X}_R) \subset \Omega(\mathbf{X})$ and $\mathbf{H}_F\mathbf{X}_R = \mathbf{X}_R$. Therefore $\mathbf{H}_F\mathbf{H}_R = \mathbf{H}_F\mathbf{X}_R(\mathbf{X}_R^T\mathbf{X}_R)^{-1}\mathbf{X}_R^T = \mathbf{H}_R$; transposing, $\mathbf{H}_R\mathbf{H}_F = \mathbf{H}_R$. Then $(\mathbf{H}_F - \mathbf{H}_R)^2 = \mathbf{H}_F - \mathbf{H}_F\mathbf{H}_R - \mathbf{H}_R\mathbf{H}_F + \mathbf{H}_R = \mathbf{H}_F - \mathbf{H}_R$, symmetric, with rank $= \text{tr}(\mathbf{H}_F) - \text{tr}(\mathbf{H}_R) = (a+b-1) - b = a-1$.

(c) $\mathbf{H}_F\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$ has entries $\mu + \tau_i + \beta_j$. $\mathbf{H}_R$ replaces each entry by its block (j) average: $\frac{1}{a}\sum_i(\mu + \tau_i + \beta_j) = \mu + \beta_j$. Hence $\xi_{ij} = \tau_i$, i.e. $\boldsymbol{\xi} = (\tau_1, \ldots, \tau_1, \ldots, \tau_a, \ldots, \tau_a)^T$ with each τ_i repeated b times, and

$$\text{E}(SS_H) = \text{tr}((\mathbf{H}_F - \mathbf{H}_R)\sigma^2\mathbf{I}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = (a-1)\sigma^2 + \boldsymbol{\xi}^T\boldsymbol{\xi} = (a-1)\sigma^2 + b\sum_i \tau_i^2.$$

(d) We use Theorem 3.11 (quadratic form in a normal vector is χ^2 if $\sigma^2\mathbf{A}$ is idempotent) and Theorem 3.12 (independence if $\mathbf{A}\sigma^2\mathbf{I}\mathbf{B} = \mathbf{0}$).

- $\mathbf{A} = (\mathbf{I} - \mathbf{H}_F)/\sigma^2$: $\sigma^2\mathbf{A} = \mathbf{I} - \mathbf{H}_F$ idempotent, rank $ab - (a+b-1)$, noncentrality $\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I} - \mathbf{H}_F)\mathbf{X}\boldsymbol{\beta}/\sigma^2 = 0$. So $SS_{E(F)}/\sigma^2 \sim \chi^2_{(a-1)(b-1)}$.
- $\mathbf{B} = (\mathbf{H}_F - \mathbf{H}_R)/\sigma^2$: idempotent (b), rank $a-1$, noncentrality $b\sum\tau_i^2/\sigma^2 = 0$ under H_0. So $SS_H/\sigma^2 \sim \chi^2_{a-1}$.
- $\mathbf{A}\sigma^2\mathbf{I}\mathbf{B} = \frac{1}{\sigma^2}(\mathbf{I} - \mathbf{H}_F)(\mathbf{H}_F - \mathbf{H}_R) = \frac{1}{\sigma^2}(\mathbf{H}_F - \mathbf{H}_R - \mathbf{H}_F + \mathbf{H}_R) = \mathbf{0}$ (using $\mathbf{H}_F\mathbf{H}_R = \mathbf{H}_R$): independent.

Ratio of independent χ^2/df: $F_0 = \frac{SS_H/(a-1)}{SS_{E(F)}/[(a-1)(b-1)]} \sim F_{a-1,(a-1)(b-1)}$ under H_0.

Solution 14

(a) Fitting the one-way model, $\mathbf{e} = (\mathbf{I} - \mathbf{H}_1)\mathbf{y}$ and

$$\mathrm{E}(\mathbf{e}) = (\mathbf{I} - \mathbf{H}_1)(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2) = \underbrace{(\mathbf{I} - \mathbf{H}_1)\mathbf{X}_1}_{=\mathbf{0}}\boldsymbol{\beta}_1 + (\mathbf{I} - \mathbf{H}_1)\mathbf{X}_2\boldsymbol{\beta}_2.$$

$\mathbf{X}_2\boldsymbol{\beta}_2$ has entries β_j; $\mathbf{H}_1$ replaces each entry by its treatment average $\frac{1}{b}\sum_j \beta_j = 0$. So $\mathbf{H}_1\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$ and $\mathrm{E}(e_{ij}) = \beta_j$: the residuals carry the block effects.

(b) $\mathrm{E}(SS_E) = \sigma^2\mathrm{tr}(\mathbf{I}-\mathbf{H}_1) + \boldsymbol{\beta}_2^T\mathbf{X}_2^T(\mathbf{I}-\mathbf{H}_1)\mathbf{X}_2\boldsymbol{\beta}_2 = \sigma^2(ab-a) + \|\mathbf{X}_2\boldsymbol{\beta}_2\|^2 = a(b-1)\sigma^2 + a\sum_j \beta_j^2$. Hence $\mathrm{E}(MS_E) = \sigma^2 + \frac{\sum_j \beta_j^2}{b-1} > \sigma^2$: biased upward.

(c) $\mathrm{E}(\bar{y}_{i\cdot}) = \mu + \tau_i + \bar{\beta} = \mu + \tau_i$, so $\hat{\tau}_i$ is still unbiased (every treatment sees every block). But MS_E is inflated, so $F_0 = MS_{\mathrm{Trt}}/MS_E$ is not $F_{a-1,a(b-1)}$ (the denominator is a noncentral χ^2): the test is conservative and loses power; CIs are too wide.

(d) SS_T is the same in both models, and $SS_E^{\text{one-way}} - SS_E^{\text{RCBD}} = SS_{\text{Block}} \geq 0$ (Solution 13(a) with roles of rows and columns swapped). Hence $R^2_{\text{RCBD}} - R^2_{\text{one-way}} = SS_{\text{Block}}/SS_T \geq 0$.

(e) $SS_T = 1.29$. RCBD: $MS_E = 0.080/9 = 0.00889$, $F_0 = \frac{0.385/3}{0.00889} = 14.44$ ($F_{0.05,3,9} = 3.86$: reject). One-way: $SS_E = 0.080 + 0.825 = 0.905$ on 12 df, $MS_E = 0.0754$, $F_0 = \frac{0.1283}{0.0754} = 1.70$ ($F_{0.05,3,12} = 3.49$: not rejected). $R^2_{\text{RCBD}} = 1 - \frac{0.080}{1.29} = 0.938$, $R^2_{\text{one-way}} = 1 - \frac{0.905}{1.29} = 0.298$; difference $0.825/1.29 = 0.640$.

Solution 15

(a) $\bar{y}_{\cdot j} = \mu + \bar{\tau} + \beta_j + \bar{\varepsilon}_{\cdot j} = \mu + \beta_j + \bar{\varepsilon}_{\cdot j}$, $\bar{y}_{\cdot\cdot} = \mu + \bar{\beta} + \bar{\varepsilon}_{\cdot\cdot}$, so $\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot} = (\beta_j - \bar{\beta}) + (\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})$, two independent zero-mean pieces:

$$\mathrm{E}(MS_{\text{Block}}) = a\Big[\mathrm{E}\frac{\sum(\beta_j - \bar{\beta})^2}{b-1} + \mathrm{E}\frac{\sum(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2}{b-1}\Big] = a\Big[\sigma_\beta^2 + \frac{\sigma^2}{a}\Big] = \sigma^2 + a\sigma_\beta^2.$$

The residual $e_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$ does not contain β_j (it cancels), so $\mathrm{E}(MS_E) = \sigma^2$ as before.

(b)

$$\mathrm{E}\big[(b-1)MS_{\text{Block}} + b(a-1)MS_E\big] = (b-1)\sigma^2 + a(b-1)\sigma_\beta^2 + b(a-1)\sigma^2 = (ab-1)\sigma^2 + a(b-1)\sigma_\beta^2.$$

Since $a(b-1) = (ab-1) - (a-1)$, this is $(ab-1)(\sigma^2 + \sigma_\beta^2) - (a-1)\sigma_\beta^2$; dividing by $ab-1$, $\mathrm{E}\hat{\sigma}^2_{\text{CRD}} = \sigma^2 + \sigma_\beta^2 \cdot \frac{a(b-1)}{ab-1} \approx \sigma^2 + \sigma_\beta^2$. (The estimator mimics the error SS a CRD would have: block SS and error SS pooled, but with the block SS "spread" over $b(a-1)$ more units.)

(c) For equal precision of a treatment mean, $\sigma^2_{\text{CRD}}/r_{\text{CRD}} = \sigma^2/r_{\text{RCBD}}$, so $r_{\text{CRD}}/r_{\text{RCBD}} = \sigma^2_{\text{CRD}}/\sigma^2 \approx RE$: a CRD needs RE times as many replicates. (A df correction $\frac{(f_b+1)(f_c+3)}{(f_b+3)(f_c+1)}$ is sometimes applied.)

(d)

$$\hat{\sigma}^2_{\text{CRD}} = \frac{3(0.275) + 4\cdot 3\,(0.00889)}{15} = \frac{0.825 + 0.1067}{15} = 0.0621, \qquad RE = \frac{0.0621}{0.00889} = 6.99.$$

Blocking is about 7 times as efficient: a CRD would need about $4 \times 6.99 \approx 28$ coupons per tip.

Solution 16

(a) $y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}$ (row i, treatment j, column k), $\sum \alpha_i = \sum \tau_j = \sum \beta_k = 0$, $\varepsilon \overset{\text{iid}}{\sim} N(0, \sigma^2)$. The rows×treatments cross product, summed over the p^2 cells, is

$$\sum_{\text{cells}} (\bar{y}_{i\cdot} - \bar{y})(\bar{y}_{\cdot j} - \bar{y}) = \sum_{i=1}^{p} (\bar{y}_{i\cdot} - \bar{y}) \sum_{\text{cells in row } i} (\bar{y}_{\cdot j} - \bar{y}) = \sum_{i} (\bar{y}_{i\cdot} - \bar{y}) \underbrace{\sum_{j=1}^{p} (\bar{y}_{\cdot j} - \bar{y})}_{0} = 0,$$

because row i contains each treatment exactly once.

(b) $\partial S/\partial \tau_j = 0$ involves the p cells with treatment j, which lie in every row and every column once: $p\hat{\mu} + \sum_i \hat{\alpha}_i + p\hat{\tau}_j + \sum_k \hat{\beta}_k = y_{\cdot j\cdot}$, so $\hat{\tau}_j = \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}$. For the EMS, $\bar{y}_{\cdot j\cdot} = \mu + \tau_j + \frac{1}{p}\sum_i \alpha_i + \frac{1}{p}\sum_k \beta_k + \bar{\varepsilon}_{\cdot j\cdot} = \mu + \tau_j + \bar{\varepsilon}_{\cdot j\cdot}$ and $\bar{y} = \mu + \bar{\varepsilon}$; thus as in the one-way case $\mathrm{E}(SS_{\text{Trt}}) = p\sum \tau_j^2 + p(p-1)\frac{\sigma^2}{p}$, i.e. $\mathrm{E}(MS_{\text{Trt}}) = \sigma^2 + \frac{p}{p-1}\sum \tau_j^2$.

(c) Row totals $32, 38, 43, 44$; column totals $38, 39, 40, 40$; treatment totals $A = 10+12+13+14 = 49$, $B = 7+9+10+10 = 36$, $C = 6+6+8+9 = 29$, $D = 9+11+12+11 = 43$; $y_{\cdots} = 157$, $\sum y^2 = 1623$, $CF = 157^2/16 = 1540.5625$.

$$\begin{aligned}
SS_T &= 1623 - 1540.5625 = 82.4375,\\
SS_{\text{Days}} &= \frac{32^2 + 38^2 + 43^2 + 44^2}{4} - CF = \frac{6253}{4} - 1540.5625 = 22.6875,\\
SS_{\text{Operators}} &= \frac{38^2 + 39^2 + 40^2 + 40^2}{4} - CF = \frac{6165}{4} - 1540.5625 = 0.6875,\\
SS_{\text{Methods}} &= \frac{49^2 + 36^2 + 29^2 + 43^2}{4} - CF = \frac{6387}{4} - 1540.5625 = 56.1875,\\
SS_E &= 82.4375 - 22.6875 - 0.6875 - 56.1875 = 2.875.
\end{aligned}$$

Source	SS	DF	MS	F_0
Methods	56.1875	3	18.729	39.09
Days	22.6875	3	7.563	
Operators	0.6875	3	0.229	
Error	2.875	6	0.479	
Total	82.4375	15		

$H_0 : \tau_A = \tau_B = \tau_C = \tau_D = 0$; $F_0 = 39.09 \sim F_{3,6}$ under H_0; $F_{0.05,3,6} = 4.76$: methods differ.

(d) $LSD = t_{0.025,6}\sqrt{2MS_E/p} = 2.447\sqrt{2(0.479)/4} = 2.447(0.4895) = 1.20$. Means $A = 12.25$, $D = 10.75$, $B = 9.00$, $C = 7.25$. Adjacent differences 1.50, 1.75, 1.75 all exceed 1.20, so all four methods differ: $A > D > B > C$.

(e) p^2-1 total df minus $3(p-1)$ for rows, columns, treatments: $(p-1)(p+1)-3(p-1) = (p-1)(p-2)$; with $p = 4$ only 6 df, so tests have low power and $p \geq 5$ (or replicated squares) is often preferred. Any interaction (e.g. day×method) has nowhere to go but the error term; because each day-method combination occurs only once, interaction is confounded with error and also biases the treatment comparisons (it is not orthogonal to them), invalidating the test.

Solution 17

(a) Counting all observations by treatments (a treatments $\times r$) and by blocks (b blocks $\times k$): $N = ar = bk$. Fix treatment 1: it lies in r blocks and meets $k-1$ other treatments in each, giving $r(k-1)$ (treatment-1, other) incidences; counted by partner, each of the $a-1$ others meets treatment 1 exactly λ times: $\lambda(a-1)$. Equate.

(b) Blocks are all k-subsets: $b = \binom{a}{k}$. A fixed treatment is in $\binom{a-1}{k-1}$ subsets: $r = \binom{a-1}{k-1}$. A fixed pair is in $\binom{a-2}{k-2}$ subsets: $\lambda = \binom{a-2}{k-2}$, the same for every pair, so it is balanced. Check: $r(k-1) = \frac{(a-1)!}{(k-1)!(a-k)!}(k-1) = (a-1)\frac{(a-2)!}{(k-2)!(a-k)!} = \lambda(a-1)$.

(c) $N = bk = 20$, $r = N/a = 4$, $\lambda = \frac{4\cdot 3}{4} = 3$. Design: block j contains all additives except j: B1 $\{2,3,4,5\}$, B2 $\{1,3,4,5\}$, B3 $\{1,2,4,5\}$, B4 $\{1,2,3,5\}$, B5 $\{1,2,3,4\}$. A pair $\{i,l\}$ is together in every block except Bi and Bl: $5-2=3=\lambda$ blocks, for every pair. (This is the unreduced design with $\binom{5}{4} = 5$.)

(d) $r = bk/a = 3$, $\lambda = \frac{3\cdot 2}{6} = 1$. Blocks: $\{1,2,4\}$, $\{2,3,5\}$, $\{3,4,6\}$, $\{4,5,7\}$, $\{5,6,1\}$, $\{6,7,2\}$, $\{7,1,3\}$. The differences within $\{1,2,4\}$ are $\pm1, \pm2, \pm3$, i.e. $\{1,6,2,5,3,4\}$ (mod 7): every nonzero residue exactly once (a *difference set*). A pair $\{x,y\}$ with $y-x \equiv d$ appears in exactly one translate (the one where d is realised), so each pair occurs exactly once: $\lambda = 1$. (This is the Fano plane / Youden square.)

(e) $r = bk/a = 24/6 = 4$, $\lambda = \frac{r(k-1)}{a-1} = \frac{12}{5}$, not an integer: no such BIBD.

Solution 18

(a)

$$\begin{aligned}
\mu &: N\hat{\mu} + r\sum_i \hat{\tau}_i + k\sum_j \hat{\beta}_j = y_{\cdot\cdot},\\
\tau_i &: r\hat{\mu} + r\hat{\tau}_i + \sum_j n_{ij}\hat{\beta}_j = y_{i\cdot},\\
\beta_j &: k\hat{\mu} + \sum_i n_{ij}\hat{\tau}_i + k\hat{\beta}_j = y_{\cdot j}. \qquad (\star)
\end{aligned}$$

(b) Multiply the τ_i equation by k: $rk\hat{\mu} + rk\hat{\tau}_i + k\sum_j n_{ij}\hat{\beta}_j = ky_{i\cdot}$. Multiply $(\star)$ by n_{ij} and sum over j: $rk\hat{\mu} + \sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l + k\sum_j n_{ij}\hat{\beta}_j = \sum_j n_{ij}y_{\cdot j}$. Subtract:

$$rk\hat{\tau}_i - \sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l = kQ_i.$$

Since $n_{ij}^2 = n_{ij}$, $\sum_j n_{ij} = r$, $\sum_j n_{ij}n_{lj} = \lambda$ $(l \neq i)$ and $\sum_{l\neq i}\hat{\tau}_l = -\hat{\tau}_i$:

$$\sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l = r\hat{\tau}_i + \lambda\sum_{l\neq i}\hat{\tau}_l = (r-\lambda)\hat{\tau}_i,$$

so $[rk - r + \lambda]\hat{\tau}_i = kQ_i$. As $r(k-1) = \lambda(a-1)$, $rk - r + \lambda = \lambda a$, giving $\hat{\tau}_i = kQ_i/(\lambda a)$. $\sum_i Q_i = \sum_i y_{i\cdot} - \frac{1}{k}\sum_j y_{\cdot j}\sum_i n_{ij} = y_{\cdot\cdot} - \frac{1}{k}\cdot k\, y_{\cdot\cdot} = 0$.

(c) $\mathrm{E}(y_{i\cdot}) = r\mu + r\tau_i + \sum_j n_{ij}\beta_j$; $\mathrm{E}(y_{\cdot j}) = k\mu + \sum_l n_{lj}\tau_l + k\beta_j$. Then

$$\begin{aligned}
\frac{1}{k}\sum_j n_{ij}\mathrm{E}(y_{\cdot j}) &= r\mu + \frac{1}{k}\sum_l\Big(\sum_j n_{ij}n_{lj}\Big)\tau_l + \sum_j n_{ij}\beta_j = r\mu + \frac{(r-\lambda)\tau_i}{k} + \sum_j n_{ij}\beta_j,\\
\mathrm{E}(Q_i) &= r\tau_i - \frac{(r-\lambda)}{k}\tau_i = \frac{r(k-1)+\lambda}{k}\tau_i = \frac{\lambda a}{k}\tau_i, \qquad \mathrm{E}(\hat{\tau}_i) = \frac{k}{\lambda a}\cdot\frac{\lambda a}{k}\tau_i = \tau_i.
\end{aligned}$$

Naive estimator: $\mathrm{E}(\bar{y}_{i\cdot}) = \mu + \tau_i + \frac{1}{r}\sum_j n_{ij}\beta_j$ and $\mathrm{E}(\bar{y}_{\cdot\cdot}) = \frac{1}{N}(N\mu + r\sum\tau_l + k\sum\beta_j) = \mu$, so $\mathrm{E}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) = \tau_i + \frac{1}{r}\sum_j n_{ij}\beta_j$: biased by the average effect of the blocks containing treatment i.

(d) $SS_E = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$ (normal equations), so $SS_{\text{Model}} = \hat{\mu}y_{\cdot\cdot} + \sum_i y_{i\cdot}\hat{\tau}_i + \sum_j y_{\cdot j}\hat{\beta}_j - \frac{y_{\cdot\cdot}^2}{N}$. From $(\star)\times y_{\cdot j}$ summed over j: $\sum_j \hat{\beta}_j y_{\cdot j} = \frac{1}{k}\sum_j y_{\cdot j}^2 - \hat{\mu}y_{\cdot\cdot} - \frac{1}{k}\sum_i\hat{\tau}_i\sum_j n_{ij}y_{\cdot j}$. Substituting and subtracting $SS_{\text{Block}} = \frac{1}{k}\sum_j y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{N}$:

$$SS_{\text{Model}} - SS_{\text{Block}} = \sum_i \hat{\tau}_i\Big(y_{i\cdot} - \frac{1}{k}\sum_j n_{ij}y_{\cdot j}\Big) = \sum_i \hat{\tau}_i Q_i = \frac{k}{\lambda a}\sum_i Q_i^2.$$

Solution 19

(a) $kQ_i = (k-1)y_{i\cdot} - \big(\sum_j n_{ij}y_{\cdot j} - y_{i\cdot}\big)$. The first piece uses the r observations of treatment i; the second is the sum of the *other* $r(k-1)$ observations in the blocks containing i. Disjoint sets $\Rightarrow$ independent, so

$$k^2\mathrm{Var}(Q_i) = (k-1)^2 r\sigma^2 + r(k-1)\sigma^2 = r(k-1)k\sigma^2 \;\Rightarrow\; \mathrm{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2.$$

(b) Use $\mathrm{Cov}(y_{i\cdot}, y_{l\cdot}) = 0$ $(i \neq l)$, $\mathrm{Cov}(y_{i\cdot}, y_{\cdot j}) = n_{ij}\sigma^2$, $\mathrm{Var}(y_{\cdot j}) = k\sigma^2$, $\mathrm{Cov}(y_{\cdot j}, y_{\cdot j'}) = 0$:

$$\mathrm{Cov}(kQ_i, kQ_l) = k^2\cdot 0 - k\sum_j n_{lj}n_{ij}\sigma^2 - k\sum_j n_{ij}n_{lj}\sigma^2 + \sum_j n_{ij}n_{lj}k\sigma^2 = -k\lambda\sigma^2,$$

so $\mathrm{Cov}(Q_i, Q_l) = -\lambda\sigma^2/k$. Consistency: $\mathrm{Var}(\sum_i Q_i) = a\frac{r(k-1)}{k}\sigma^2 - a(a-1)\frac{\lambda}{k}\sigma^2 = \frac{a\sigma^2}{k}[r(k-1) - \lambda(a-1)] = 0$, as it must since $\sum Q_i \equiv 0$.

(c)

$$\mathrm{Var}(\hat\tau_i) = \frac{k^2}{\lambda^2 a^2}\cdot\frac{r(k-1)}{k}\sigma^2 = \frac{k\,\lambda(a-1)}{\lambda^2 a^2}\sigma^2 = \frac{k(a-1)}{\lambda a^2}\sigma^2,$$
$$\mathrm{Var}(\hat\tau_i - \hat\tau_l) = \frac{k^2}{\lambda^2 a^2}\Big[\frac{2r(k-1)}{k} + \frac{2\lambda}{k}\Big]\sigma^2 = \frac{2k}{\lambda^2 a^2}[\lambda(a-1)+\lambda]\sigma^2 = \frac{2k}{\lambda a}\sigma^2.$$

(d) RCBD with r blocks: $\mathrm{Var}(\hat\tau_i - \hat\tau_l) = 2\sigma^2/r$. Efficiency $E = \frac{2\sigma^2/r}{2k\sigma^2/(\lambda a)} = \frac{\lambda a}{rk} = \frac{a(k-1)}{k(a-1)} = \frac{1-1/k}{1-1/a} < 1$ iff $k < a$. Catalyst: $E = \frac{2\cdot 4}{3\cdot 3} = 0.889$; $a = 7$, $k = 3$, $\lambda = 1$, $r = 3$: $E = \frac{7}{9} = 0.778$. (The loss is the price of incomplete blocks; it is usually more than offset by the smaller σ^2 in small blocks.)

Solution 20

(a) Incidence: trt 1 in blocks 1,2,3; trt 2 in 1,2,4; trt 3 in 1,3,4; trt 4 in 2,3,4. Totals $y_{1\cdot} = 37$, $y_{2\cdot} = 48$, $y_{3\cdot} = 34$, $y_{4\cdot} = 52$; $y_{\cdot 1} = 37$, $y_{\cdot 2} = 48$, $y_{\cdot 3} = 37$, $y_{\cdot 4} = 49$; $y_{\cdot\cdot} = 171$.

$$Q_1 = 37 - \tfrac{1}{3}(37+48+37) = -3.667, \qquad Q_2 = 48 - \tfrac{1}{3}(37+48+49) = 3.333,$$
$$Q_3 = 34 - \tfrac{1}{3}(37+37+49) = -7.000, \qquad Q_4 = 52 - \tfrac{1}{3}(48+37+49) = 7.333,$$

sum $= 0$ ✓.

(b) $\sum y^2 = 2531$, $CF = 171^2/12 = 2436.75$.

$$\begin{aligned} SS_T &= 2531 - 2436.75 = 94.25, \\ SS_{\text{Block}} &= \frac{37^2+48^2+37^2+49^2}{3} - 2436.75 = \frac{7443}{3} - 2436.75 = 44.25, \\ SS_{\text{Trt(adj)}} &= \frac{3\,(3.667^2+3.333^2+7^2+7.333^2)}{2\cdot 4} = \frac{3(127.33)}{8} = 47.75, \\ SS_E &= 94.25 - 44.25 - 47.75 = 2.25. \end{aligned}$$

Source	SS	DF	MS	F_0
Treatments (adj.)	47.75	3	15.92	35.37
Blocks	44.25	3	14.75	
Error	2.25	5	0.45	
Total	94.25	11		

$F_0 = 35.37 > F_{0.05,3,5} = 5.41$: treatments differ.

(c) $\hat\tau_i = 3Q_i/8$: $(-1.375,\ 1.250,\ -2.625,\ 2.750)$. $se(\hat\tau_i - \hat\tau_l) = \sqrt{2kMS_E/(\lambda a)} = \sqrt{6(0.45)/8} = 0.581$. Treatments 3 vs 4: $t_0 = \frac{-2.625-2.750}{0.581} = -9.25$, $|t_0| > t_{0.025,5} = 2.571$: significantly different.

(d) Unadjusted: $\frac{37^2+48^2+34^2+52^2}{3} - CF = \frac{7533}{3} - 2436.75 = 74.25$. Then the blocks adjusted for treatments get $94.25 - 2.25 - 74.25 = 17.75$. Sequential (type I) SS depend on the order because treatment and block columns are not orthogonal in an incomplete design; but SS_E is the residual SS of the same full model (same column space) in both orders, 2.25. The correct test of treatments uses the SS *adjusted* for blocks, i.e. `block` first.

Solution 21

(a) $S = \sum_{ijk} \left(y_{ijk} - \mu - \tau_i - \beta_j - (\tau\beta)_{ij}\right)^2$. Normal equations (using the zero-sum constraints on the estimates):

$$\begin{aligned}
\partial/\partial\mu : abn\hat{\mu} = y_{\cdots} \;&\Rightarrow\; \hat{\mu} = \bar{y}_{\cdots},\\
\partial/\partial\tau_i : bn\hat{\mu} + bn\hat{\tau}_i = y_{i\cdot\cdot} \;&\Rightarrow\; \hat{\tau}_i = \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots},\\
\partial/\partial\beta_j : an\hat{\mu} + an\hat{\beta}_j = y_{\cdot j\cdot} \;&\Rightarrow\; \hat{\beta}_j = \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots},\\
\partial/\partial(\tau\beta)_{ij} : n(\hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \widehat{(\tau\beta)}_{ij}) = y_{ij\cdot} \;&\Rightarrow\; \widehat{(\tau\beta)}_{ij} = \bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}.
\end{aligned}$$

Adding, $\hat{\mu}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j + \widehat{(\tau\beta)}_{ij} = \bar{y}_{ij\cdot}$: the full model fits each cell by its mean.

(b) $SS_{AB} = n\sum\sum \widehat{(\tau\beta)}_{ij}^2$ involves ab quantities restricted by $\sum_i \widehat{(\tau\beta)}_{ij} = 0$ (b equations) and $\sum_j \widehat{(\tau\beta)}_{ij} = 0$ (a equations). These $a+b$ equations have one dependency (summing all row equations equals summing all column equations: both say the grand sum is 0), so $a+b-1$ are independent; df $= ab-(a+b-1) = (a-1)(b-1)$. Equivalently, df(cells)$-$df(A)$-$df(B) $= (ab-1)-(a-1)-(b-1)$.

(c) $\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots} = (\tau\beta)_{ij} + D_{ij}$, $D_{ij} = \bar{\varepsilon}_{ij\cdot} - \bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdot j\cdot} + \bar{\varepsilon}_{\cdots}$ (main effects cancel; the interaction terms reduce to $(\tau\beta)_{ij}$ by the zero-sum constraints). The $\bar{\varepsilon}_{ij\cdot}$ are ab iid $N(0, \sigma^2/n)$ variables and D_{ij} is their RCBD residual, so by Solution 12(c) (with σ^2 replaced by σ^2/n) $\mathrm{E}\sum\sum D_{ij}^2 = (a-1)(b-1)\sigma^2/n$. The cross term has zero mean. Thus

$$\mathrm{E}(SS_{AB}) = n\sum\sum(\tau\beta)_{ij}^2 + (a-1)(b-1)\sigma^2, \qquad \mathrm{E}(MS_{AB}) = \sigma^2 + \frac{n\sum\sum(\tau\beta)_{ij}^2}{(a-1)(b-1)}.$$

(d) $\bar{y}_{\cdots} = \frac{1}{b}\sum_{j'} \bar{y}_{\cdot j'\cdot}$, the $\bar{y}_{\cdot j'\cdot}$ being independent with variance $\sigma^2/(an)$. So $\mathrm{Cov}(\bar{y}_{\cdots}, \bar{y}_{\cdot j\cdot}) = \frac{1}{b}\cdot\frac{\sigma^2}{an} = \frac{\sigma^2}{abn} = \mathrm{Var}(\bar{y}_{\cdots})$, and

$$\mathrm{Cov}(\hat{\mu}, \hat{\beta}_j) = \mathrm{Cov}(\bar{y}_{\cdots}, \bar{y}_{\cdot j\cdot}) - \mathrm{Var}(\bar{y}_{\cdots}) = 0.$$

Jointly normal + uncorrelated $\Rightarrow$ independent. Similarly ($i \neq i'$):

$$\mathrm{Cov}(\hat{\tau}_i, \hat{\tau}_{i'}) = \mathrm{Cov}(\bar{y}_{i\cdot\cdot}, \bar{y}_{i'\cdot\cdot}) - \mathrm{Cov}(\bar{y}_{i\cdot\cdot}, \bar{y}_{\cdots}) - \mathrm{Cov}(\bar{y}_{\cdots}, \bar{y}_{i'\cdot\cdot}) + \mathrm{Var}(\bar{y}_{\cdots}) = 0 - \tfrac{\sigma^2}{abn} - \tfrac{\sigma^2}{abn} + \tfrac{\sigma^2}{abn} = -\frac{\sigma^2}{abn},$$

$$\mathrm{Var}(\hat{\tau}_i) = \frac{\sigma^2}{bn} - 2\frac{\sigma^2}{abn} + \frac{\sigma^2}{abn} = \frac{(a-1)\sigma^2}{abn}.$$

(Check: $\mathrm{Var}(\sum_i \hat{\tau}_i) = a\frac{(a-1)\sigma^2}{abn} - a(a-1)\frac{\sigma^2}{abn} = 0$.)

(e) In terms of cell means, $\widehat{(\tau\beta)}_{ij} = \sum_{i'',j'}(\delta_{ii''} - \frac{1}{a})(\delta_{jj'} - \frac{1}{b})\bar{y}_{i''j'\cdot}$ and $\hat{\tau}_{i'} = \sum_{i'',j'}(\delta_{i'i''} - \frac{1}{a})\frac{1}{b}\,\bar{y}_{i''j'\cdot}$. With independent cell means of variance σ^2/n,

$$\mathrm{Cov}(\widehat{(\tau\beta)}_{ij}, \hat{\tau}_{i'}) = \frac{\sigma^2}{nb}\sum_{i''}\Big(\delta_{ii''} - \frac{1}{a}\Big)\Big(\delta_{i'i''} - \frac{1}{a}\Big)\underbrace{\sum_{j'}\Big(\delta_{jj'} - \frac{1}{b}\Big)}_{=1-b/b=0} = 0.$$

Solution 22

(a)

Model	Error SS	df
M1	$\sum\sum\sum(y_{ijk} - \bar{y}_{ij\cdot})^2$	$ab(n-1)$
M2	$\sum\sum\sum(y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2$	$abn - a - b + 1$
M3	$\sum\sum(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$	$(a-1)(b-1)$

For M2, $y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots} = (y_{ijk} - \bar{y}_{ij\cdot}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})$, and the cross term vanishes because $\sum_k(y_{ijk} - \bar{y}_{ij\cdot}) = 0$. Hence $SS_E^{(2)} = SS_E^{(1)} + SS_{AB}$, and the df add: $ab(n-1) + (a-1)(b-1) = abn - a - b + 1$.

(b) The numerator of all three is $SS_A/(a-1)$, with $SS_A = bn\sum(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2$ (M3: $n = 1$, cell value y_{ij}). If $y_{ijk} \approx \bar{y}_{ij\cdot} \approx y_{ij1}$:

- M1: $SS_E^{(1)} \approx 0$, so $F^{(1)}$ is huge.
- M2: $SS_E^{(2)} \approx SS_{AB} = n\sum\sum(\bar{y}_{ij\cdot} - \cdots)^2 \approx n\,SS_E^{(3)}$ and $SS_A^{(2)} = bn\sum(\cdot)^2 \approx n\,SS_A^{(3)}$, so

$$F^{(2)} \approx \frac{n\,SS_A^{(3)}/(a-1)}{n\,SS_E^{(3)}/(abn - a - b + 1)} = F^{(3)} \cdot \frac{abn - a - b + 1}{(a-1)(b-1)}.$$

Thus $F^{(1)} \gg F^{(2)} \geq F^{(3)}$. Replication without real within-cell variation inflates $F^{(2)}$ artificially through the larger df; M1 identifies whether the "residual" of M2 is interaction or pure error.

(c) $\mathrm{E}(SS_E^{(2)}) = \mathrm{E}(SS_E^{(1)}) + \mathrm{E}(SS_{AB}) = ab(n-1)\sigma^2 + (a-1)(b-1)\sigma^2 + n\sum\sum(\tau\beta)^2$, so

$$\mathrm{E}(MS_E^{(2)}) = \sigma^2 + \frac{n\sum\sum(\tau\beta)_{ij}^2}{abn - a - b + 1} > \sigma^2.$$

The denominator is inflated: main-effect F tests become conservative (power loss), and, more seriously, main effects may be meaningless when interaction is present.

(d) The additive model pools interaction into error: $SS_E = 9.00 + 0.50 = 9.50$ with $6 + 2 = 8$ df, $MS_E = 9.50/8 = 1.1875$. New F: A: $75.00/1.1875 = 63.16$ ($F_{0.05,1,8} = 5.32$); B: $(32.17/2)/1.1875 = 16.083/1.1875 = 13.54$ ($F_{0.05,2,8} = 4.46$). Both remain significant; pooling was harmless because $F_{AB} = 0.167$ was tiny.

Solution 23

(a) Since $\sum_i \tau_i = \sum_j \beta_j = 0$, $\sum_j \gamma\tau_i\beta_j = \gamma\tau_i \sum_j \beta_j = 0$ and $\sum_i \sum_j \gamma\tau_i\beta_j = 0$. Hence

$$\bar{y}_{\cdot\cdot} = \mu + \bar{\varepsilon}_{\cdot\cdot}, \qquad \bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}, \qquad \bar{y}_{\cdot j} = \mu + \beta_j + \bar{\varepsilon}_{\cdot j},$$

giving $\mathrm{E}\bar{y}_{\cdot\cdot} = \mu$, $\mathrm{E}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) = \tau_i$, $\mathrm{E}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}) = \beta_j$ for any γ.

(b) With $S_{ij} = y_{ij} - \mu - \tau_i - \beta_j - \gamma\tau_i\beta_j$, $\partial S/\partial\gamma = -2\sum\sum \tau_i\beta_j S_{ij} = 0$:

$$\sum\sum \tau_i\beta_j y_{ij} - \mu \underbrace{\sum\sum \tau_i\beta_j}_{0} - \underbrace{\sum_i \tau_i^2 \sum_j \beta_j}_{0} - \underbrace{\sum_i \tau_i \sum_j \beta_j^2}_{0} - \gamma \sum_i \tau_i^2 \sum_j \beta_j^2 = 0,$$

so $\tilde{\gamma} = \frac{\sum\sum \tau_i\beta_j y_{ij}}{\sum \tau_i^2 \sum \beta_j^2}$ and, plugging in estimates, $\hat{\gamma} = \frac{\sum\sum \hat{\tau}_i\hat{\beta}_j y_{ij}}{\sum \hat{\tau}_i^2 \sum \hat{\beta}_j^2}$. The SS explained by the term $\hat{\gamma}\hat{\tau}_i\hat{\beta}_j$ is

$$SS_N = \sum\sum (\hat{\gamma}\hat{\tau}_i\hat{\beta}_j)^2 = \hat{\gamma}^2 \sum \hat{\tau}_i^2 \sum \hat{\beta}_j^2 = \frac{(\sum\sum \hat{\tau}_i\hat{\beta}_j y_{ij})^2}{\sum \hat{\tau}_i^2 \sum \hat{\beta}_j^2},$$

and since $SS_A = b\sum \hat{\tau}_i^2$, $SS_B = a\sum \hat{\beta}_j^2$, $\sum \hat{\tau}_i^2 \sum \hat{\beta}_j^2 = SS_A SS_B/(ab)$.

(c) $\sum_j c_{ij} = \hat{\tau}_i \sum_j \hat{\beta}_j = 0$, $\sum_i c_{ij} = \hat{\beta}_j \sum_i \hat{\tau}_i = 0$. With $y_{ij} = \bar{y}_{\cdot\cdot} + \hat{\tau}_i + \hat{\beta}_j + e_{ij}$:

$$\sum\sum c_{ij} y_{ij} = \bar{y}_{\cdot\cdot} \sum\sum c_{ij} + \sum_i \hat{\tau}_i \sum_j c_{ij} + \sum_j \hat{\beta}_j \sum_i c_{ij} + \sum\sum c_{ij} e_{ij} = \sum\sum c_{ij} e_{ij}.$$

(d) Under H_0 the additive model holds and $\mathbf{y}$ is normal; the residual vector $\mathbf{e}$ is a linear function of $\mathbf{y}$ uncorrelated with (hence independent of) the row and column means, which determine the c_{ij}. Conditional on $(\hat{\tau}, \hat{\beta})$, $\sum\sum c_{ij} e_{ij} = \sum\sum c_{ij} y_{ij}$ (by (c)), and since $\mathbf{c}$ lies in the residual space (zero row/column sums), $\sum\sum c_{ij} y_{ij}$ given the margins is $N(0, \sigma^2 \sum\sum c_{ij}^2)$, i.e. $N(0, \sigma^2 \sum \hat{\tau}_i^2 \sum \hat{\beta}_j^2)$. Hence

$$\frac{SS_N}{\sigma^2} = \frac{(\sum\sum c_{ij} y_{ij})^2}{\sigma^2 \sum\sum c_{ij}^2} \;\Big|\; (\hat{\tau}, \hat{\beta}) \;\sim \chi^2_1,$$

which does not depend on the conditioning variables, so it holds unconditionally. Now $Q = SS_{\mathrm{Res}}/\sigma^2 \sim \chi^2_{(a-1)(b-1)}$, $Q_1 = SS_N/\sigma^2 \sim \chi^2_1$, and $Q_2 = Q - Q_1 = SS_E/\sigma^2 \geq 0$ (it is a residual SS after projecting out one more direction). By the Cochran-type lemma $Q_2 \sim \chi^2_{(a-1)(b-1)-1}$ independent of Q_1, hence

$$F_0 = \frac{SS_N/1}{SS_E/[(a-1)(b-1)-1]} \overset{H_0}{\sim} F_{1,(a-1)(b-1)-1}.$$

Solution 24

(a) Row totals $48, 67, 91$; column totals $43, 52, 64, 47$; $y_{\cdot\cdot} = 206$, $\bar{y}_{\cdot\cdot} = 17.167$, $\sum y^2 = 3856$, $CF = 206^2/12 = 3536.33$.

$$\hat{\tau} = (16,\ 16.75,\ 22.75) - 17.167 = (-5.167,\ -0.417,\ 5.583), \qquad \sum \hat{\tau}_i^2 = 58.042,$$

$$\hat{\beta} = (14.333,\ 17.333,\ 21.333,\ 15.667) - 17.167 = (-2.833,\ 0.167,\ 4.167,\ -1.500), \qquad \sum \hat{\beta}_j^2 = 27.667.$$

$SS_A = 4(58.042) = 232.17$, $SS_B = 3(27.667) = 83.00$, $SS_T = 3856 - 3536.33 = 319.67$, $SS_{\text{Res}} = 319.67 - 232.17 - 83.00 = 4.50$.

(b) $\sum\sum \hat{\tau}_i \hat{\beta}_j y_{ij} = \sum_i \hat{\tau}_i \big(\sum_j \hat{\beta}_j y_{ij}\big)$. Row-wise $\sum_j \hat{\beta}_j y_{ij}$: row 1: $-2.833(10) + 0.167(12) + 4.167(15) - 1.5(11) = -28.33 + 2 + 62.5 - 16.5 = 19.67$; row 2: $-39.67 + 2.83 + 87.5 - 22.5 = 28.17$; row 3: $-53.83 + 3.83 + 116.67 - 31.5 = 35.17$. Then $-5.167(19.67) - 0.417(28.17) + 5.583(35.17) = -101.6 - 11.7 + 196.4 = 83.0$.

$$SS_N = \frac{83.0^2}{58.042 \times 27.667} = \frac{6889}{1605.8} = 4.29, \qquad SS_E = 4.50 - 4.29 = 0.21 \text{ (5 df)}, \qquad MS_E = 0.042,$$

$F_0 = 4.29/0.042 = 102.2 > F_{0.05,1,5} = 6.61$: strong evidence of non-additivity.

(c) If $y_{ij} = \mu\alpha_i\beta_j$ then $\log y_{ij} = \log\mu + \log\alpha_i + \log\beta_j$ is additive. Writing $\alpha_i = 1 + a_i$, $\beta_j = 1 + b_j$ with small a_i, b_j: $\mu\alpha_i\beta_j = \mu + \mu a_i + \mu b_j + \mu a_i b_j$. Identify $\tau_i \approx \mu a_i$, $\beta_j \approx \mu b_j$; then the product term is $\mu a_i b_j = \frac{1}{\mu}\tau_i\beta_j$, exactly Tukey's form with $\gamma = 1/\mu$. So Tukey's test is designed to detect precisely the curvature produced by a multiplicative (or more generally a power-transformable) structure; its estimate even suggests a power transformation $y^{1-\hat{\gamma}\bar{y}}$.

(d) On the log scale $SS_N = 1.85 \times 10^{-5}$, $SS_E = 1.54 \times 10^{-4} - 1.85 \times 10^{-5} = 1.36 \times 10^{-4}$ on 5 df: $F_0 = \frac{1.85\times10^{-5}}{1.36\times10^{-4}/5} = 0.68 < 6.61$. No non-additivity remains: analyse $\log y$ with the additive model.

Solution 25

(a) Cell totals: A_1: $21, 24, 28$; A_2: $30, 35, 46$. $y_{1\cdot\cdot} = 73$, $y_{2\cdot\cdot} = 111$; $y_{\cdot 1\cdot} = 51$, $y_{\cdot 2\cdot} = 59$, $y_{\cdot 3\cdot} = 74$; $y_{\cdots} = 184$, $\sum y^2 = 3030$, $CF = 184^2/12 = 2821.33$.

$$
\begin{aligned}
SS_T &= 3030 - 2821.33 = 208.67,\\
SS_A &= \frac{73^2 + 111^2}{6} - 2821.33 = \frac{17650}{6} - 2821.33 = 120.33,\\
SS_B &= \frac{51^2 + 59^2 + 74^2}{4} - 2821.33 = \frac{11558}{4} - 2821.33 = 68.17,\\
SS_{\text{Cells}} &= \frac{21^2 + 24^2 + 28^2 + 30^2 + 35^2 + 46^2}{2} - 2821.33\\
&= \frac{441 + 576 + 784 + 900 + 1225 + 2116}{2} - 2821.33\\
&= \frac{6042}{2} - 2821.33 = 199.67,\\
SS_{AB} &= SS_{\text{Cells}} - SS_A - SS_B = 199.67 - 120.33 - 68.17 = 11.17,\\
SS_E &= SS_T - SS_{\text{Cells}} = 208.67 - 199.67 = 9.00.
\end{aligned}
$$

Source	SS	DF	MS	F_0
A	120.33	1	120.33	80.22
B	68.17	2	34.08	22.72
AB	11.17	2	5.58	3.72
Error	9.00	6	1.50	
Total	208.67	11		

$F_{0.05,1,6} = 5.99$, $F_{0.05,2,6} = 5.14$: A and B significant; AB not significant at 5% ($P \approx 0.09$).

(b) Cell means: A_1: 10.5, 12.0, 14.0; A_2: 15.0, 17.5, 23.0. Both profiles increase with B; the A_2 profile is steeper (gap 4.5, 5.5, 9.0), a mild divergence — the suggested interaction, not significant with only 6 error df.

(c) $LSD = t_{0.025,6}\sqrt{2MS_E/n} = 2.447\sqrt{1.5} = 2.447(1.225) = 3.00$. Within A_1: $|B_1 - B_2| = 1.5$, $|B_2 - B_3| = 2.0$ n.s.; $|B_1 - B_3| = 3.5 > 3.00$ *. Within A_2: $|B_1 - B_2| = 2.5$ n.s.; $|B_2 - B_3| = 5.5$ *, $|B_1 - B_3| = 8.0$ *. So under A_2 level B_3 is clearly best; under A_1 only B_1 vs B_3 differ.

(d) $\widehat{(\tau\beta)}_{23} = \bar{y}_{23\cdot} - \bar{y}_{2\cdot\cdot} - \bar{y}_{\cdot 3\cdot} + \bar{y}_{\cdots} = 23.0 - 18.5 - 18.5 + 15.333 = 1.333$. Its coefficient on cell mean $\bar{y}_{i'j'\cdot}$ is $(\delta_{2i'} - \frac{1}{a})(\delta_{3j'} - \frac{1}{b})$, so

$$
\text{Var} = \frac{\sigma^2}{n}\sum_{i'}\Big(\delta_{2i'} - \frac{1}{a}\Big)^2 \sum_{j'}\Big(\delta_{3j'} - \frac{1}{b}\Big)^2 = \frac{\sigma^2}{n}\cdot\frac{a-1}{a}\cdot\frac{b-1}{b} = \frac{\sigma^2}{2}\cdot\frac{1}{2}\cdot\frac{2}{3} = \frac{\sigma^2}{6},
$$

estimated by $1.5/6 = 0.25$ (se 0.5).

Solution 26

(a) Simple effects: at B^-, $\frac{a-(1)}{n}$; at B^+, $\frac{ab-b}{n}$. Average: $\frac{1}{2n}[a-(1)+ab-b]$. Also $\bar{y}_{A^+} = \frac{a+ab}{2n}$, $\bar{y}_{A^-} = \frac{(1)+b}{2n}$, difference $\frac{ab+a-b-(1)}{2n}$.

(b) With $a=b=2$, $SS_A = 2n\sum_i(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2$. $\bar{y}_{A^+} - \bar{y}_{\cdots} = \frac{2(ab+a)-(ab+a+b+(1))}{4n} = \frac{ab+a-b-(1)}{4n}$, and $\bar{y}_{A^-} - \bar{y}_{\cdots}$ is its negative. Hence $SS_A = 2n \cdot 2 \cdot \frac{[ab+a-b-(1)]^2}{16n^2} = \frac{[ab+a-b-(1)]^2}{4n}$.

(c) With cell totals $S_\ell = n\bar{y}_\ell$, $\mathrm{Var}(\sum c_\ell S_\ell) = n^2 \sum c_\ell^2 \frac{\sigma^2}{n} = n\sigma^2 \sum c_\ell^2$. In the effects model $ab = n(\mu + A_2 + B_2 + (AB)_{22}) + \varepsilon_{22\cdot}$ etc., so

$$\mathrm{E}(\mathrm{Contrast}_A) = n\big[2A_2 - 2A_1 + (AB)_{22} + (AB)_{21} - (AB)_{12} - (AB)_{11}\big] = n(2A_2 - 2A_1),$$

the interaction terms cancelling because $\sum_i (AB)_{ij} = 0$. Under $H_0 : A_1 = A_2 = 0$, $\mathrm{E}(\mathrm{Contrast}_A) = 0$, so $t_0 = \frac{\mathrm{Contrast}_A}{\sqrt{4n\,MS_E}} \sim t_{4(n-1)}$ and $t_0^2 = \frac{\mathrm{Contrast}_A^2/(4n)}{MS_E} = \frac{SS_A}{MS_E} \sim F_{1,4(n-1)}$.

(d) Coefficient vectors on $((1), a, b, ab)$: $A = (-1,1,-1,1)$, $B = (-1,-1,1,1)$, $AB = (1,-1,-1,1)$. Inner products: $A \cdot B = 1-1-1+1 = 0$, $A \cdot AB = -1-1+1+1 = 0$, $B \cdot AB = -1+1-1+1 = 0$. Together with $(1,1,1,1)/2$ they form an orthogonal basis of $\mathbb{R}^4$, so (as in Solution 7(c)) $SS_A + SS_B + SS_{AB}$ equals the between-cells SS $\sum_\ell S_\ell^2/n - y_{\cdots}^2/(4n)$.

Solution 27

(a) A main-effect column has 2^{k-1} plus and 2^{k-1} minus signs. Any non-identity column is a product $\prod_{i\in E} x_i$ ($x_i = \pm 1$); summing over the full factorial, $\sum \prod_{i\in E} x_i = \prod_{i\in E} \left(\sum_{x_i=\pm 1} x_i\right) \cdot 2^{k-|E|} = 0$. Two distinct columns $E \neq F$: their product is the column $E \triangle F$ (since $x_i^2 = 1$), which is a non-identity column, so its sum, i.e. their inner product, is 0.

(b) Effect $= \bar{y}_+ - \bar{y}_-$, each average over $2^{k-1}n$ observations: $\frac{\sum_+ S_\ell - \sum_- S_\ell}{2^{k-1}n} = \frac{\text{Contrast}}{2^{k-1}n}$. $SS = \frac{\text{Contrast}^2}{n\sum c_\ell^2} = \frac{\text{Contrast}^2}{2^k n}$. Var(Contrast) $= n\sigma^2 \cdot 2^k$, so Var(Effect) $= \frac{2^k n\sigma^2}{2^{2(k-1)}n^2} = \frac{\sigma^2}{2^{k-2}n}$.

(c) Columns of $\mathbf{X}$ are $\mathbf{1}$ and the sign columns, each replicated n times. By (a), off-diagonal entries of $\mathbf{X}^T\mathbf{X}$ are 0 and diagonal ones $2^k n$: $\mathbf{X}^T\mathbf{X} = 2^k n\mathbf{I}$. Thus $\hat{\beta}_E = \frac{1}{2^k n}\sum(\text{signs})\, y = \frac{\text{Contrast}_E}{2^k n} = \frac{1}{2}\text{Effect}_E$. Regression SS of the term: $\hat{\beta}_E^2 \cdot \|\mathbf{x}_E\|^2 = \frac{\text{Contrast}^2}{(2^k n)^2} 2^k n = \frac{\text{Contrast}^2}{2^k n} = SS_E$ (the ANOVA SS); orthogonality makes these SS order-free.

(d) $\frac{\text{Effect}-\text{E(Effect)}}{\sqrt{MS_E/(2^{k-2}n)}} \sim t_{2^k(n-1)}$ (normal numerator, independent χ^2 denominator), giving Effect $\pm t_{0.025,2^k(n-1)}\sqrt{MS_E/(2^{k-2}n)}$.

Solution 28

(a) Totals: $(1) = 53$, $a = 75$, $b = 69$, $ab = 102$, $c = 89$, $ac = 77$, $bc = 110$, $abc = 80$ (sum 655).

$$\begin{aligned}
\text{Contrast}_A &= 75 + 102 + 77 + 80 - 53 - 69 - 89 - 110 = 334 - 321 = 13,\\
\text{Contrast}_B &= 69 + 102 + 110 + 80 - 53 - 75 - 89 - 77 = 361 - 294 = 67,\\
\text{Contrast}_C &= 89 + 77 + 110 + 80 - 53 - 75 - 69 - 102 = 356 - 299 = 57,\\
\text{Contrast}_{AB} &= 53 + 102 + 89 + 80 - 75 - 69 - 77 - 110 = 324 - 331 = -7,\\
\text{Contrast}_{AC} &= 53 + 69 + 77 + 80 - 75 - 102 - 89 - 110 = 279 - 376 = -97,\\
\text{Contrast}_{BC} &= 53 + 75 + 110 + 80 - 69 - 102 - 89 - 77 = 318 - 337 = -19,\\
\text{Contrast}_{ABC} &= 75 + 69 + 89 + 80 - 53 - 102 - 77 - 110 = 313 - 342 = -29.
\end{aligned}$$

Effect = Contrast/8, SS = Contrast^2/16:

Term	Effect	SS	$F_0 = SS/MS_E$
A	1.625	10.56	0.44
B	8.375	280.56	11.78*
C	7.125	203.06	8.53*
AB	−0.875	3.06	0.13
AC	−12.125	588.06	24.70*
BC	−2.375	22.56	0.95
ABC	−3.625	52.56	2.21

(b) $\sum y^2 = 28165$, $CF = 655^2/16 = 26814.06$, $SS_T = 1350.94$; $\sum SS_{\text{effects}} = 1160.44$; $SS_E = 190.50$ on $2^3(2-1) = 8$ df, $MS_E = 23.81$. $F_{0.05,1,8} = 5.32$: B, C and AC are significant.

(c) Keep A for hierarchy (AC is in the model): $\bar{y}_{\cdot\cdot} = 655/16 = 40.94$,

$$\hat{y} = 40.94 + \frac{1.625}{2}x_1 + \frac{8.375}{2}x_2 + \frac{7.125}{2}x_3 - \frac{12.125}{2}x_1x_3 = 40.94 + 0.81x_1 + 4.19x_2 + 3.56x_3 - 6.06x_1x_3.$$

(d) $A \times C$ means (each over 4 observations): $A^-C^- = \frac{(1)+b}{4} = \frac{53+69}{4} = 30.50$; $A^+C^- = \frac{a+ab}{4} = 44.25$; $A^-C^+ = \frac{c+bc}{4} = 49.75$; $A^+C^+ = \frac{ac+abc}{4} = 39.25$. At low C, raising A increases the response by 13.75; at high C it decreases it by 10.50. $AC = \frac{1}{2}(-10.50 - 13.75) = -12.125$ ✓. The small main effect of A hides two large opposite simple effects.

Solution 29

(a) Model $y = \mu + \text{effects} + \delta_{\text{block}} + \varepsilon$, block $1 = \{(1), ab, ac, bc\}$, block $2 = \{a, b, c, abc\}$. Each contrast is $\sum_\ell c_\ell S_\ell$; the block effects contribute $n(\delta_1 \sum_{\ell \in B_1} c_\ell + \delta_2 \sum_{\ell \in B_2} c_\ell)$. For every column other than ABC, each block contains two $+$ and two $-$ (check e.g. A on block 1: $(1)-, ab+, ac+, bc-$), so both sums are 0. For ABC, block 2 is exactly its $+$ set: $\mathrm{E}(\text{Contrast}_{ABC}) = (\text{true ABC part}) + 4n(\delta_2 - \delta_1)$.

(b) The column $E = \{i : \alpha_i = 1\}$ has sign $\prod_{i \in E}(2x_i - 1)$, and $2x_i - 1 = (-1)^{1-x_i}$. So the sign is $(-1)^{\sum_i \alpha_i (1-x_i)} = (-1)^{|E| - L}$, which for fixed E depends only on $L \bmod 2$. Thus $\{L \equiv 0\}$ and $\{L \equiv 1\}$ are exactly the two sign classes of the confounded column.

(c) L is linear mod 2 in the exponent vectors, and multiplying treatment combinations adds exponent vectors mod 2. If $L(u) \equiv L(v) \equiv 0$ then $L(uv) \equiv 0$: closure ((1) is the identity and each element is its own inverse, so it is a subgroup). If $L(w) \equiv 1$ then $L(wu) \equiv 1$ for all u in the principal block and $u \mapsto wu$ is one-to-one, so $w\cdot$(principal block) is the whole other block (a coset).

(d) Generalized interaction $ABC \cdot BCD = AB^2C^2D = AD$; three effects ABC, BCD, AD are confounded (3 block df). With $L_1 = x_1 + x_2 + x_3$, $L_2 = x_2 + x_3 + x_4$ (mod 2):

$(L_1, L_2) = (0,0)$	$(1,0)$	$(0,1)$	$(1,1)$
(1)	a	d	b
bc	abc	ab	c
abd	bd	ac	ad
acd	cd	bcd	$abcd$

Check closure of the principal block: $bc \cdot abd = ab^2cd = acd$ ✓; other blocks are cosets, e.g. $a \cdot \{(1), bc, abd, acd\} = \{a, abc, bd, cd\}$. All 12 remaining effects are estimable free of blocks.

Solution 30

(a)

Rep	Confounded	$L \equiv 0$ (block 1)	$L \equiv 1$ (block 2)
1	AB: $L = x_1 + x_2$	$(1), ab, c, abc$	a, b, ac, bc
2	AC: $L = x_1 + x_3$	$(1), b, ac, abc$	a, ab, c, bc
3	BC: $L = x_2 + x_3$	$(1), a, bc, abc$	b, ab, c, ac
4	ABC: $L = x_1 + x_2 + x_3$	$(1), ab, ac, bc$	a, b, c, abc

(b)

Source	DF
Replicates	3
Blocks within replicates	4
A, B, C	1 each
AB (reps 2,3,4), AC (reps 1,3,4), BC (reps 1,2,4), ABC (reps 1,2,3)	1 each
Error	17
Total	31

$3+4+7+17 = 31 = 32-1$. The 7 block df (8 blocks -1) $= 3$ replicate $+4$ blocks-within-replicates.

(c) AB is estimated only from replicates 2–4 (in rep 1 its contrast equals the block difference): $\text{Contrast}_{AB}^{(2,3,4)}$ sums totals over those 3 replicates, $SS_{AB} = \frac{(\text{Contrast}^{(2,3,4)})^2}{3 \cdot 2^3}$, and $\widehat{AB} = \text{Contrast}^{(2,3,4)}/(3 \cdot 4)$ with $\text{Var} = \frac{\sigma^2}{2^{k-2} \cdot 3} = \frac{\sigma^2}{6}$. A main effect uses all 4 replicates: $\text{Var}(\hat{A}) = \frac{\sigma^2}{2 \cdot 4} = \frac{\sigma^2}{8}$. Relative information $= \frac{1/(\sigma^2/6)}{1/(\sigma^2/8)} = \frac{6}{8} = \frac{3}{4}$.

(d) Complete confounding of ABC: ABC is lost entirely (no test), all other effects have full information (variance $\sigma^2/8$). Partial confounding: every interaction is estimable and testable, each with 3/4 information. Prefer partial confounding when no interaction can be assumed negligible; prefer complete confounding of ABC when it is surely negligible.

Solution 31

(a) *Rules.* (1) $\mathbf{a}^T\boldsymbol{\beta} = \sum_k a_k\beta_k$, so $\partial(\mathbf{a}^T\boldsymbol{\beta})/\partial\beta_k = a_k$: the gradient is $\mathbf{a}$. (2) $\boldsymbol{\beta}^T\mathbf{A}\boldsymbol{\beta} = \sum_k\sum_l A_{kl}\beta_k\beta_l$; differentiating w.r.t. β_m gives $\sum_l A_{ml}\beta_l + \sum_k A_{km}\beta_k = (\mathbf{A}\boldsymbol{\beta} + \mathbf{A}^T\boldsymbol{\beta})_m$, i.e. $2\mathbf{A}\boldsymbol{\beta}$ for symmetric $\mathbf{A}$. *Normal equations.* $L(\boldsymbol{\beta}) = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$ (the two cross terms are equal scalars), so $\partial L/\partial\boldsymbol{\beta} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0} \Rightarrow \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$. *Minimum.* For any $\boldsymbol{\beta}$, $L(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2$ (the cross term $2(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T\mathbf{X}^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = 0$ by the normal equations), so $L(\boldsymbol{\beta}) \geq L(\hat{\boldsymbol{\beta}})$ with equality iff $\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{0} \iff \boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ (full rank).

(b) $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}$, so $\mathrm{E}\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}$ and

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\sigma^2\mathbf{I}\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

Also

$$\begin{aligned} SS_E &= \mathbf{y}^T\mathbf{y} - 2\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} + \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} \\ &= \mathbf{y}^T\mathbf{y} - 2\hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} + \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}. \end{aligned}$$

(c)

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\sigma^2\mathbf{I})(\mathbf{I} - \mathbf{H})^T = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^T - \mathbf{X}^T\mathbf{H}) = \mathbf{0},$$

because $\mathbf{X}^T\mathbf{H} = (\mathbf{H}\mathbf{X})^T = \mathbf{X}^T$. Under normality $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathbf{e}$ and hence $\hat{\boldsymbol{\beta}} \perp\!\!\!\perp SS_E = \mathbf{e}^T\mathbf{e}$. $\mathrm{E}(SS_E) = \sigma^2\mathrm{tr}(\mathbf{I} - \mathbf{H}) + 0 = (n-p)\sigma^2$, and $SS_E/\sigma^2 \sim \chi^2_{n-p}$ (Theorem 3.11). Therefore

$$t_0 = \frac{(\hat{\beta}_j - \beta_j)/\sqrt{\sigma^2 C_{jj}}}{\sqrt{SS_E/[\sigma^2(n-p)]}} = \frac{\hat{\beta}_j - \beta_j}{\sqrt{MS_E\, C_{jj}}} \sim t_{n-p}.$$

(d) $\hat{y}_0 = \mathbf{x}_0^T\hat{\boldsymbol{\beta}}$ has $\mathrm{E} = \mathbf{x}_0^T\boldsymbol{\beta}$, $\mathrm{Var} = \sigma^2\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$; with MS_E for σ^2 the pivot is t_{n-p} and

$$\hat{y}_0 \pm t_{\alpha/2,n-p}\sqrt{MS_E\,\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0}.$$

For a new observation $y_0 = \mathbf{x}_0^T\boldsymbol{\beta} + \varepsilon_0$ independent of the data, $\mathrm{Var}(y_0 - \hat{y}_0) = \sigma^2 + \sigma^2\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$, giving the prediction interval $\hat{y}_0 \pm t_{\alpha/2,n-p}\sqrt{MS_E[1 + \mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0]}$. The extra "1" is the variance of the new error, which does not shrink with n.

Solution 32

(a) $\sum x = 21$, $\sum x^2 = 91$, $\sum y = 42.3$, $\sum xy = 183.4$, $\sum y^2 = 369.75$.

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} 6 & 21 \\ 21 & 91 \end{pmatrix},\ \det = 546 - 441 = 105, \quad \mathbf{X}^T\mathbf{y} = \begin{pmatrix} 42.3 \\ 183.4 \end{pmatrix},$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{105}\begin{pmatrix} 91 & -21 \\ -21 & 6 \end{pmatrix} = \begin{pmatrix} 0.8667 & -0.2 \\ -0.2 & 0.05714 \end{pmatrix},$$

$$\hat{\beta}_0 = 0.8667(42.3) - 0.2(183.4) = 36.66 - 36.68 = -0.02,$$

$$\hat{\beta}_1 = -0.2(42.3) + 0.05714(183.4) = -8.46 + 10.48 = 2.02.$$

Check: $S_{xx} = 91 - 21^2/6 = 17.5$, $S_{xy} = 183.4 - 21(42.3)/6 = 183.4 - 148.05 = 35.35$, $\hat{\beta}_1 = 35.35/17.5 = 2.02$, $\hat{\beta}_0 = 7.05 - 2.02(3.5) = -0.02$ ✓.

(b) $S_{yy} = 369.75 - 42.3^2/6 = 369.75 - 298.215 = 71.535$; $SS_R = \hat{\beta}_1 S_{xy} = 2.02(35.35) = 71.407$; $SS_E = 71.535 - 71.407 = 0.128$ (4 df), $MS_E = 0.032$.

Source	SS	DF	MS	F_0
Regression	71.407	1	71.407	2231.5
Error	0.128	4	0.032	
Total	71.535	5		

$F_0 \gg F_{0.05,1,4} = 7.71$. Slope: $se(\hat{\beta}_1) = \sqrt{MS_E/S_{xx}} = \sqrt{0.032/17.5} = 0.04276$, $t_0 = 2.02/0.04276 = 47.24$, $t_0^2 = 2231.5 = F_0$ ✓ ($C_{11} = 0.05714 = 1/S_{xx}$).

(c) $t_{0.025,4} = 2.776$. β_1: $2.02 \pm 2.776(0.04276) = 2.02 \pm 0.119 = [1.901, 2.139]$. $x_0 = 3.5 = \bar{x}$: $\hat{y}_0 = 7.05$, $se = \sqrt{0.032(\frac{1}{6} + 0)} = 0.0730$, CI $7.05 \pm 0.203 = [6.85, 7.25]$. $x_0 = 6$: $\hat{y}_0 = 12.10$, $se = \sqrt{0.032(\frac{1}{6} + \frac{2.5^2}{17.5})} = \sqrt{0.032(0.5238)} = 0.1295$, CI $12.10 \pm 0.359 = [11.74, 12.46]$. Since $\mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}$, uncertainty grows with distance from $\bar{x}$ (slope error is magnified).

(d) $\sum_\ell \mathrm{Var}(\hat{y}_\ell) = \sigma^2 \mathrm{tr}(\mathbf{H}) = \sigma^2 \mathrm{tr}((\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}) = 2\sigma^2$. Directly: $\sum_\ell \sigma^2[\frac{1}{6} + \frac{(x_\ell - \bar{x})^2}{S_{xx}}] = \sigma^2(1 + 1) = 2\sigma^2$.

Solution 33

(a) $\mathrm{Var}(y_{ij}) = \sigma_\tau^2 + \sigma^2$; $\mathrm{Cov}(y_{ij}, y_{ij'}) = \mathrm{E}[(\tau_i + \varepsilon_{ij})(\tau_i + \varepsilon_{ij'})] = \sigma_\tau^2$ $(j \neq j')$; $\mathrm{Cov}(y_{ij}, y_{i'j'}) = 0$ $(i \neq i')$. Hence

$$\boldsymbol{\Sigma} = \mathbf{I}_a \otimes (\sigma^2 \mathbf{I}_n + \sigma_\tau^2 \mathbf{J}_n) = \sigma^2 \mathbf{I}_N + \sigma_\tau^2 (\mathbf{I}_a \otimes \mathbf{J}_n).$$

(b) $\mathbf{A} = \frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n - \frac{1}{N}\mathbf{J}_N$: $\mathbf{Ay}$ has entries $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, and $\mathbf{A}$ is symmetric idempotent, so $\mathbf{y}^T\mathbf{Ay} = \|\mathbf{Ay}\|^2 = n\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_A$.

(c) Using $(\mathbf{P} \otimes \mathbf{Q})(\mathbf{R} \otimes \mathbf{S}) = \mathbf{PR} \otimes \mathbf{QS}$ and $\mathbf{J}_n^2 = n\mathbf{J}_n$:

$$\tfrac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n)\boldsymbol{\Sigma} = \mathbf{I}_a \otimes \tfrac{1}{n}(\sigma^2 \mathbf{J}_n + n\sigma_\tau^2 \mathbf{J}_n) = (\sigma^2 + n\sigma_\tau^2)\tfrac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n),$$
$$\tfrac{1}{N}\mathbf{J}_N\boldsymbol{\Sigma} = \tfrac{1}{N}(\mathbf{J}_a \otimes \mathbf{J}_n)(\mathbf{I}_a \otimes (\sigma^2 \mathbf{I}_n + \sigma_\tau^2 \mathbf{J}_n)) = \tfrac{1}{N}\mathbf{J}_a \otimes (\sigma^2 + n\sigma_\tau^2)\mathbf{J}_n = (\sigma^2 + n\sigma_\tau^2)\tfrac{1}{N}\mathbf{J}_N.$$

So $\mathbf{A}\boldsymbol{\Sigma} = (\sigma^2 + n\sigma_\tau^2)\mathbf{A}$ and $\frac{\mathbf{A}\boldsymbol{\Sigma}}{\sigma^2 + n\sigma_\tau^2} = \mathbf{A}$, idempotent. For $\mathbf{y} \sim N(\mu\mathbf{1}, \boldsymbol{\Sigma})$, $\mathbf{y}^T\mathbf{My} \sim \chi^2_{\mathrm{rank}\mathbf{M}}(\mu^2 \mathbf{1}^T\mathbf{M1})$ when $\mathbf{M}\boldsymbol{\Sigma}$ is idempotent; take $\mathbf{M} = \mathbf{A}/(\sigma^2 + n\sigma_\tau^2)$: rank $= \mathrm{tr}\mathbf{A} = a - 1$, noncentrality 0 (since $\mathbf{A1} = \mathbf{0}$). Hence $\frac{SS_A}{\sigma^2 + n\sigma_\tau^2} \sim \chi^2_{a-1}$.

(d) $\mathrm{E}(\mathbf{y}^T\mathbf{Ay}) = \mathrm{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \mu^2 \mathbf{1}^T\mathbf{A1} = (\sigma^2 + n\sigma_\tau^2)(a-1)$. With $\mathbf{B} = \mathbf{I}_N - \frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n$ ($SS_E = \mathbf{y}^T\mathbf{By}$): $\mathbf{B}\boldsymbol{\Sigma} = \sigma^2\mathbf{B} + \sigma_\tau^2(\mathbf{I}_a \otimes \mathbf{J}_n - \mathbf{I}_a \otimes \frac{1}{n}\mathbf{J}_n^2) = \sigma^2\mathbf{B}$, so $\mathrm{E}(SS_E) = \sigma^2 \mathrm{tr}\mathbf{B} = \sigma^2(N - a)$. Equating: $\hat{\sigma}^2 = MS_E$, $\hat{\sigma}_\tau^2 = (MS_A - MS_E)/n$.

(e) $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B} = (\sigma^2 + n\sigma_\tau^2)\mathbf{AB}$ and $\mathbf{AB} = \mathbf{A} - \mathbf{A}(\frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n) = \mathbf{A} - \mathbf{A} = \mathbf{0}$ (since $\frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n$ fixes both pieces of $\mathbf{A}$). Hence $SS_A \perp\!\!\!\perp SS_E$. With $SS_E/\sigma^2 \sim \chi^2_{N-a}$ ($\mathbf{B}\boldsymbol{\Sigma}/\sigma^2 = \mathbf{B}$ idempotent),

$$\frac{MS_A/(\sigma^2 + n\sigma_\tau^2)}{MS_E/\sigma^2} \sim F_{a-1,N-a} \;\Rightarrow\; F_0 = \frac{MS_A}{MS_E} \sim \frac{\sigma^2 + n\sigma_\tau^2}{\sigma^2} F_{a-1,N-a},$$

which is $F_{a-1,N-a}$ under $H_0 : \sigma_\tau^2 = 0$ and stochastically larger otherwise.

Solution 34

(a) Totals $225, 211, 229, 219, 237$; $y_{\cdot\cdot} = 1121$; $\sum y^2 = 83929$; $CF = 1121^2/15 = 83776.07$.

$$SS_T = 83929 - 83776.07 = 152.93,$$
$$SS_{\text{Batch}} = \frac{225^2 + 211^2 + 229^2 + 219^2 + 237^2}{3} - CF = \frac{251717}{3} - 83776.07 = 129.60,$$
$$SS_E = 152.93 - 129.60 = 23.33.$$

Source	SS	DF	MS	F_0
Batches	129.60	4	32.40	13.89
Error	23.33	10	2.333	
Total	152.93	14		

$H_0 : \sigma_\tau^2 = 0$; $F_0 \sim F_{4,10}$ under H_0; $13.89 > F_{0.05,4,10} = 3.48$: significant batch-to-batch variability.

(b) $\hat\sigma^2 = 2.333$, $\hat\sigma_\tau^2 = \frac{32.40-2.333}{3} = 10.02$, $\hat\rho = \frac{10.02}{10.02+2.333} = 0.811$: about 81% of the total variance is between batches.

(c) $\chi^2_{0.025,10} = 20.48$, $\chi^2_{0.975,10} = 3.247$: $\sigma^2 \in [\frac{23.33}{20.48}, \frac{23.33}{3.247}] = [1.14, 7.19]$. For ρ: $\mathcal{A} = \frac{MS_A/(\sigma^2+n\sigma_\tau^2)}{MS_E/\sigma^2} = \frac{F^*}{1+n\theta} \sim F_{4,10}$, $F^* = 13.886$, $F_{0.025,4,10} = 4.468$, $F_{0.975,4,10} = 0.1131$. Inverting $0.1131 \le \frac{F^*}{1+3\theta} \le 4.468$:

$$L = \frac{1}{3}\Big(\frac{13.886}{4.468} - 1\Big) = 0.703, \qquad U = \frac{1}{3}\Big(\frac{13.886}{0.1131} - 1\Big) = 40.6,$$
$$\frac{L}{1+L} = 0.413 \le \rho \le \frac{U}{1+U} = 0.976.$$

(d) $\hat\sigma_\tau^2 < 0 \iff MS_A < MS_E \iff \frac{MS_A/(\sigma^2+n\sigma_\tau^2)}{MS_E/\sigma^2} < \frac{\sigma^2}{\sigma^2+n\sigma_\tau^2} = \frac{1}{1+n\theta}$; the left side is $F_{a-1,N-a}$. With $a-1 = 4$, $N-a = 10$, $n = 3$: $\theta = 0$: $P(F < 1) = 0.548$; $\theta = 0.5$: $P(F < 0.4) = 0.196$; $\theta = 1$: $P(F < 0.25) = 0.097$. Negative estimates are common when σ_τ^2 is small.

Solution 35

(a) Let $x = \mathrm{Cov}((\tau\beta)_{ij}, (\tau\beta)_{i'j})$, the same for all $i \neq i'$. From $\sum_i (\tau\beta)_{ij} = 0$: $0 = \mathrm{Var}(\sum_i (\tau\beta)_{ij}) = a\frac{a-1}{a}\sigma^2_{\tau\beta} + a(a-1)x \Rightarrow x = -\sigma^2_{\tau\beta}/a$.

(b) MS_A. $\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots} = \tau_i + \overline{(\tau\beta)}_{i\cdot} + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})$ ($\bar{\beta}$ cancels, $\overline{(\tau\beta)}_{\cdot\cdot} = 0$). $\overline{(\tau\beta)}_{i\cdot} = \frac{1}{b}\sum_j (\tau\beta)_{ij}$ averages over independent columns: $\mathrm{Var} = \frac{a-1}{ab}\sigma^2_{\tau\beta}$. Hence

$$\mathrm{E}(MS_A) = \frac{bn}{a-1}\Big[\sum \tau_i^2 + a \cdot \frac{a-1}{ab}\sigma^2_{\tau\beta} + (a-1)\frac{\sigma^2}{bn}\Big] = \sigma^2 + n\sigma^2_{\tau\beta} + \frac{bn\sum \tau_i^2}{a-1}.$$

MS_B. $\bar{y}_{\cdot j\cdot} = \mu + \beta_j + \overline{(\tau\beta)}_{\cdot j} + \bar{\varepsilon}_{\cdot j\cdot}$ with $\overline{(\tau\beta)}_{\cdot j} = 0$ by the restriction: $\mathrm{E}(MS_B) = an[\sigma^2_\beta + \frac{\sigma^2}{an}] = \sigma^2 + an\sigma^2_\beta$. MS_{AB}. $\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots} = \big((\tau\beta)_{ij} - \overline{(\tau\beta)}_{i\cdot}\big) + (\varepsilon\text{-part})$; for fixed i the $(\tau\beta)_{ij}$, $j = 1..b$, are iid with variance $\frac{a-1}{a}\sigma^2_{\tau\beta}$, so $\mathrm{E}\sum_i \sum_j (\cdot)^2 = a(b-1)\frac{a-1}{a}\sigma^2_{\tau\beta}$; with the error part $(a-1)(b-1)\sigma^2/n$: $\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma^2_{\tau\beta}$.

(c) Unrestricted: $\overline{(\alpha\gamma)}_{i\cdot} - \overline{(\alpha\gamma)}_{\cdot\cdot}$ with iid $\overline{(\alpha\gamma)}_{i\cdot}$ of variance $\sigma^2_{\alpha\gamma}/b$ gives $\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\alpha\gamma} + \frac{bn\sum \alpha_i^2}{a-1}$; and $\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots} = (\gamma_j - \bar{\gamma}) + (\overline{(\alpha\gamma)}_{\cdot j} - \overline{(\alpha\gamma)}_{\cdot\cdot}) + (\bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}_{\cdots})$ with $\mathrm{Var}(\overline{(\alpha\gamma)}_{\cdot j}) = \sigma^2_{\alpha\gamma}/a$ gives $\mathrm{E}(MS_B) = an\big[\sigma^2_\gamma + \frac{\sigma^2_{\alpha\gamma}}{a} + \frac{\sigma^2}{an}\big] = \sigma^2 + n\sigma^2_{\alpha\gamma} + an\sigma^2_\gamma$.

(d) Restricted: $F_0 = MS_B/MS_E$, $\hat{\sigma}^2_\beta = (MS_B - MS_E)/(an)$. Unrestricted: $F_0 = MS_B/MS_{AB}$, $\hat{\sigma}^2_\gamma = (MS_B - MS_{AB})/(an)$. (In both, A is tested against MS_{AB}.) EMS algorithm (restricted; columns i:F,a; j:R,b; k:R,n):

Term	i	j	k	EMS
τ_i	0	b	n	$\sigma^2 + n\sigma^2_{\tau\beta} + bn\sum \tau_i^2/(a-1)$
β_j	a	1	n	$\sigma^2 + an\sigma^2_\beta$
$(\tau\beta)_{ij}$	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta}$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

For β_j, covering column j leaves $(\tau\beta)_{ij}$ with $0 \cdot n = 0$: the restriction makes the interaction drop out — matching (b).

Solution 36

(a) Columns i (F, a), j (R, b), k (R, c), l (R, n). Rule: dead subscript $\to$ 1; live subscript $\to$ 0 if fixed, 1 if random; absent $\to$ number of levels.

Term	i	j	k	l	EMS
τ_i	0	b	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + bn\sigma^2_{\tau\gamma} + bcn\frac{\sum \tau_i^2}{a-1}$
β_j	a	1	c	n	$\sigma^2 + an\sigma^2_{\beta\gamma} + acn\sigma^2_{\beta}$
γ_k	a	b	1	n	$\sigma^2 + an\sigma^2_{\beta\gamma} + abn\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	0	1	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta}$
$(\tau\gamma)_{ik}$	0	b	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + bn\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	a	1	1	n	$\sigma^2 + an\sigma^2_{\beta\gamma}$
$(\tau\beta\gamma)_{ijk}$	0	1	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

E.g. β_j: cover column j; rows containing j: β_j $(a \cdot c \cdot n)$, $(\tau\beta)_{ij}$ $(0 \cdot c \cdot n = 0)$, $(\beta\gamma)_{jk}$ $(a \cdot 1 \cdot n)$, $(\tau\beta\gamma)_{ijk}$ $(0 \cdot 1 \cdot n = 0)$, ε (1). For τ_i: rows with i: τ_i (bcn), $(\tau\beta)_{ij}$ $(1 \cdot c \cdot n)$, $(\tau\gamma)_{ik}$ $(b \cdot 1 \cdot n)$, $(\tau\beta\gamma)$ $(1 \cdot 1 \cdot n)$, ε.

(b) Exact tests (numerator and denominator EMS differ only by the tested component): B: MS_B/MS_{BC}; C: MS_C/MS_{BC}; AB: MS_{AB}/MS_{ABC}; AC: MS_{AC}/MS_{ABC}; BC: MS_{BC}/MS_E; ABC: MS_{ABC}/MS_E.

(c) Under $H_0 : \tau_i = 0$, $\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + bn\sigma^2_{\tau\gamma} = \mathrm{E}(MS_{AB}) + \mathrm{E}(MS_{AC}) - \mathrm{E}(MS_{ABC})$. Quasi-$F$:

$$F_0 = \frac{MS_A}{MS_{AB} + MS_{AC} - MS_{ABC}}, \qquad \nu = \frac{(MS_{AB} + MS_{AC} - MS_{ABC})^2}{\frac{MS_{AB}^2}{(a-1)(b-1)} + \frac{MS_{AC}^2}{(a-1)(c-1)} + \frac{MS_{ABC}^2}{(a-1)(b-1)(c-1)}},$$

$F_0 \approx F_{a-1,\nu}$. (Alternatively $F_0' = \frac{MS_A + MS_{ABC}}{MS_{AB} + MS_{AC}}$, which avoids a negative denominator, with Satterthwaite df for both parts.)

(d) A, C random, B fixed (Ch. 9 table): $\mathrm{E}(MS_A) = \sigma^2 + bn\sigma^2_{\tau\gamma} + bcn\sigma^2_{\tau}$, $\mathrm{E}(MS_C) = \sigma^2 + bn\sigma^2_{\tau\gamma} + abn\sigma^2_{\gamma}$, $\mathrm{E}(MS_{AC}) = \sigma^2 + bn\sigma^2_{\tau\gamma}$, $\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta}$, $\mathrm{E}(MS_{BC}) = \sigma^2 + n\sigma^2_{\tau\beta\gamma} + an\sigma^2_{\beta\gamma}$, $\mathrm{E}(MS_{ABC}) = \sigma^2 + n\sigma^2_{\tau\beta\gamma}$. Exact tests: A: MS_A/MS_{AC}; C: MS_C/MS_{AC}; AC: MS_{AC}/MS_E; AB: MS_{AB}/MS_{ABC}; BC: MS_{BC}/MS_{ABC}; ABC: MS_{ABC}/MS_E. For $H_0 : \beta_j = 0$: $F_0 = \frac{MS_B}{MS_{AB} + MS_{BC} - MS_{ABC}}$ (approximate, Satterthwaite df).

Solution 37

(a) $y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$, $i = 1,2$, $j = 1,2,3$, $k = 1,2$; $\sum \tau_i = 0$, $\beta_{j(i)} \overset{\text{iid}}{\sim} N(0, \sigma^2_\beta)$, $\varepsilon \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Decompose $y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot}) + (y_{ijk} - \bar{y}_{ij\cdot})$; cross terms vanish since $\sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot}) = 0$ and $\sum_k (y_{ijk} - \bar{y}_{ij\cdot}) = 0$: $SS_T = SS_A + SS_{B(A)} + SS_E$, df $11 = 1 + 4 + 6$ (generally $(a-1) + a(b-1) + ab(n-1)$).

(b) $\bar{y}_{i\cdot\cdot} = \mu + \tau_i + \bar{\beta}_{\cdot(i)} + \bar{\varepsilon}_{i\cdot\cdot}$ with $\mathrm{Var}(\bar{\beta}_{\cdot(i)}) = \sigma^2_\beta / b$ independent across i:

$$\mathrm{E}(MS_A) = \frac{bn}{a-1}\Big[\sum \tau_i^2 + (a-1)\Big(\frac{\sigma^2_\beta}{b} + \frac{\sigma^2}{bn}\Big)\Big] = \sigma^2 + n\sigma^2_\beta + \frac{bn\sum \tau_i^2}{a-1}.$$

$\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} = (\beta_{j(i)} - \bar{\beta}_{\cdot(i)}) + (\bar{\varepsilon}_{ij\cdot} - \bar{\varepsilon}_{i\cdot\cdot})$: $\mathrm{E}(MS_{B(A)}) = \frac{n}{a(b-1)} \cdot a(b-1)(\sigma^2_\beta + \frac{\sigma^2}{n}) = \sigma^2 + n\sigma^2_\beta$.

(c) Batch totals $25, 29, 22 \,|\, 35, 30, 39$; supplier totals $76, 104$; $y_{\cdots} = 180$; $\sum y^2 = 2804$; $CF = 2700$.

$$SS_T = 104.00, \quad SS_A = \frac{76^2 + 104^2}{6} - 2700 = \frac{16592}{6} - 2700 = 65.33,$$
$$SS_{B(A)} = \frac{25^2 + 29^2 + 22^2 + 35^2 + 30^2 + 39^2}{2} - \frac{16592}{6} = \frac{5596}{2} - 2765.33 = 32.67,$$
$$SS_E = 104 - 65.33 - 32.67 = 6.00.$$

Source	SS	DF	MS	EMS	F_0
Suppliers	65.33	1	65.33	$\sigma^2 + 2\sigma^2_\beta + 6\sum \tau_i^2$	$65.33/8.17 = 8.00$
Batches(S)	32.67	4	8.17	$\sigma^2 + 2\sigma^2_\beta$	$8.17/1.00 = 8.17$
Error	6.00	6	1.00	σ^2	
Total	104.00	11			

$F_{0.05,1,4} = 7.71$: suppliers differ (barely). $F_{0.05,4,6} = 4.53$: batch variance significant. $\hat{\sigma}^2 = 1.00$, $\hat{\sigma}^2_\beta = \frac{8.17 - 1.00}{2} = 3.58$, $\hat{\tau} = (76/6 - 15,\ 104/6 - 15) = (-2.33,\ 2.33)$.

(d) Crossed analysis (batch labels 1–3 matched across suppliers, totals $60, 59, 61$):

$$SS_B = \frac{60^2 + 59^2 + 61^2}{4} - 2700 = 0.50, \qquad SS_{AB} = 32.67 - 0.50 = 32.17.$$

Indeed $SS_{B(A)} = SS_B + SS_{AB}$ and $a(b-1) = (b-1) + (a-1)(b-1)$. The crossed "batch" main effect is meaningless (batch 1 of supplier 1 has nothing to do with batch 1 of supplier 2), and testing suppliers against the wrong error would be invalid.

(e) $y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$, plants $\tau_i \sim N(0, \sigma^2_\tau)$, leaves within plants $\beta_{j(i)} \sim N(0, \sigma^2_\beta)$, both random, leaves nested.

Source	DF	EMS	F_0
Plants	2	$\sigma^2 + n\sigma^2_\beta + 2n\sigma^2_\tau$	$MS_A / MS_{B(A)}$
Leaves(Plants)	3	$\sigma^2 + n\sigma^2_\beta$	$MS_{B(A)} / MS_E$
Error	$6(n-1)$	σ^2	
Total	$6n - 1$		

Solution 38

(a) Within each block, a level of A is applied to a whole plot (e.g. one batch / one oven setting), and the three B levels are randomized to subplots *within* that whole plot. Randomization is therefore restricted: B is randomized within whole plots, A among whole plots within blocks. Model II: $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_k + (\beta\gamma)_{jk} + \varepsilon_{ijk}$, i block ($\tau_i \sim N(0, \sigma_\tau^2)$), j factor A ($\sum \beta_j = 0$), $(\tau\beta)_{ij} \sim N(0, \sigma_{\tau\beta}^2)$ whole-plot error, k factor B ($\sum \gamma_k = 0$), $\sum_j (\beta\gamma)_{jk} = \sum_k (\beta\gamma)_{jk} = 0$, $\varepsilon_{ijk} \sim N(0, \sigma^2)$ subplot error.

(b) $(r-1) + (a-1) + (r-1)(a-1) = ra - 1$ (the whole-plot table) and $(b-1) + (a-1)(b-1) + a(r-1)(b-1) = (b-1)[1 + (a-1) + a(r-1)] = (b-1)ar$. Total $ra - 1 + rab - ra = rab - 1$.

Term	R i (r)	F j (a)	F k (b)	R l (1)	EMS
τ_i	1	a	b	1	$\sigma^2 + ab\sigma_\tau^2$
β_j	r	0	b	1	$\sigma^2 + b\sigma_{\tau\beta}^2 + rb\sum \beta_j^2/(a-1)$
$(\tau\beta)_{ij}$	1	0	b	1	$\sigma^2 + b\sigma_{\tau\beta}^2$
γ_k	r	a	0	1	$\sigma^2 + ra\sum \gamma_k^2/(b-1)$
$(\beta\gamma)_{jk}$	r	0	0	1	$\sigma^2 + r\sum\sum (\beta\gamma)^2/[(a-1)(b-1)]$
ε_{ijk}	1	1	1	1	σ^2

(c) Block totals $94, 104$; A totals $89, 109$; B totals $50, 67, 81$; block×A totals $41, 53, 48, 56$; A×B totals $23, 30, 36, 27, 37, 45$; $y_{\cdots} = 198$, $\sum y^2 = 3438$, $CF = 3267$.

$$SS_T = 171.00, \quad SS_{\text{Block}} = \frac{94^2 + 104^2}{6} - CF = 8.33, \quad SS_A = \frac{89^2 + 109^2}{6} - CF = 33.33,$$

$$SS_{E(w)} = \frac{41^2 + 53^2 + 48^2 + 56^2}{3} - CF - 8.33 - 33.33 = 3310 - 3267 - 41.67 = 1.33,$$

$$SS_B = \frac{50^2 + 67^2 + 81^2}{4} - CF = 3387.5 - 3267 = 120.50,$$

$$SS_{AB} = \frac{23^2 + 30^2 + 36^2 + 27^2 + 37^2 + 45^2}{2} - CF - 33.33 - 120.50 = 3424 - 3267 - 153.83 = 3.17,$$

$$SS_{E(s)} = 171 - 8.33 - 33.33 - 1.33 - 120.50 - 3.17 = 4.33.$$

Source	SS	DF	MS	F_0
Blocks	8.33	1	8.33	
A	33.33	1	33.33	$33.33/1.33 = 25.0$
Whole-plot error	1.33	1	1.33	
B	120.50	2	60.25	$60.25/1.083 = 55.6$
AB	3.17	2	1.58	$1.58/1.083 = 1.46$
Subplot error	4.33	4	1.083	
Total	171.00	11		

A: $F_{0.05,1,1} = 161.4$, not significant despite $F_0 = 25$. B: $F_{0.05,2,4} = 6.94$, significant. AB: not significant.

(d) The whole-plot error has only $(r-1)(a-1) = 1$ df, so the critical value is enormous and the A comparison has almost no power, although the A difference (89 vs 109) looks large. More blocks (replicates) are needed for the whole-plot factor; B and AB are estimated precisely against the subplot error. Paint study (4 mixes whole plot, 3 methods subplot, 3 days): days 2, mix 3, whole-plot error 6, method 2, mix×method 6, subplot error $4 \cdot 2 \cdot 2 = 16$, total 35; tests: mix $MS_A/MS_{E(w)}$, method and interaction against $MS_{E(s)}$.

Solution 39

(a) $S = \sum\sum (y_{ij} - \mu - \tau_i - \beta(x_{ij} - \bar{x}_{\cdot\cdot}))^2$. ∂_μ: $N\hat{\mu} = y_{\cdot\cdot}$ (since $\sum \hat{\tau}_i = 0$ and $\sum\sum(x_{ij} - \bar{x}_{\cdot\cdot}) = 0$) $\Rightarrow \hat{\mu} = \bar{y}_{\cdot\cdot}$. ∂_{τ_i}: $n\hat{\mu} + n\hat{\tau}_i + \hat{\beta}n(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}) = y_{i\cdot} \Rightarrow \hat{\tau}_i = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) - \hat{\beta}(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})$. ∂_β: $\sum\sum y_{ij}(x_{ij} - \bar{x}_{\cdot\cdot}) - \sum_i \hat{\tau}_i n(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}) - \hat{\beta}S_{xx} = 0$, i.e. $S_{xy} = n\sum_i \hat{\tau}_i(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}) + \hat{\beta}S_{xx}$. Substituting $\hat{\tau}_i$: $S_{xy} = T_{xy} - \hat{\beta}T_{xx} + \hat{\beta}S_{xx}$, so $\hat{\beta} = \frac{S_{xy}-T_{xy}}{S_{xx}-T_{xx}} = \frac{E_{xy}}{E_{xx}}$.

(b) $\hat{y}_{ij} = \bar{y}_{i\cdot} + \hat{\beta}(x_{ij} - \bar{x}_{i\cdot})$, $e_{ij} = (y_{ij} - \bar{y}_{i\cdot}) - \hat{\beta}(x_{ij} - \bar{x}_{i\cdot})$: $SS_E = E_{yy} - 2\hat{\beta}E_{xy} + \hat{\beta}^2 E_{xx} = E_{yy} - E_{xy}^2/E_{xx}$. Parameters μ, $a-1$ τ's, β: df $an - (a+1) = a(n-1) - 1$.

(c) $\hat{\beta} = \sum\sum(x_{ij} - \bar{x}_{i\cdot})y_{ij}/E_{xx}$, so $\mathrm{Var}\hat{\beta} = \sigma^2 \sum\sum(x_{ij} - \bar{x}_{i\cdot})^2/E_{xx}^2 = \sigma^2/E_{xx}$. $\mathrm{Cov}(\bar{y}_{i\cdot}, \hat{\beta}) = \frac{1}{nE_{xx}}\sum_j (x_{ij} - \bar{x}_{i\cdot})\sigma^2 = 0$. Then $\hat{\tau}_i - \hat{\tau}_l = (\bar{y}_{i\cdot} - \bar{y}_{l\cdot}) - \hat{\beta}(\bar{x}_{i\cdot} - \bar{x}_{l\cdot})$ and

$$\mathrm{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{2\sigma^2}{n} + (\bar{x}_{i\cdot} - \bar{x}_{l\cdot})^2 \frac{\sigma^2}{E_{xx}}.$$

(d) $H_0 : \tau_i = 0$: reduced model $y = \mu + \beta(x - \bar{x}) + \varepsilon$, $SS_E(R) = S_{yy} - S_{xy}^2/S_{xx}$ ($an - 2$ df); $F_0 = \frac{[SS_E(R) - SS_E(F)]/(a-1)}{SS_E(F)/[a(n-1)-1]} \sim F_{a-1,a(n-1)-1}$. $H_0 : \beta = 0$: reduced one-way model, $SS_E(R) = E_{yy}$; $SS_E(R) - SS_E(F) = E_{xy}^2/E_{xx}$, so $F_0 = \frac{E_{xy}^2/E_{xx}}{MS_E} = \frac{\hat{\beta}^2}{MS_E/E_{xx}} = t_0^2$ with $t_0 = \hat{\beta}/se(\hat{\beta})$.

(e) ANACOVA assumes x is a concomitant variable unaffected by treatments. If treatments change x, then adjusting to a common $\bar{x}_{\cdot\cdot}$ removes part of the very treatment effect being studied ($\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}$ would itself reflect treatment), and the adjusted means become meaningless or biased.

Solution 40

(a) $\sum x = 362$, $\sum y = 603$, $\sum x^2 = 8998$, $\sum y^2 = 24587$, $\sum xy = 14835$; machine totals x: $126, 130, 106$; y: $207, 216, 180$.

$$S_{xx} = 8998 - \tfrac{362^2}{15} = 261.73, \quad S_{yy} = 24587 - \tfrac{603^2}{15} = 346.40, \quad S_{xy} = 14835 - \tfrac{362\cdot 603}{15} = 282.60,$$
$$T_{xx} = \tfrac{44012}{5} - 8736.27 = 66.13, \quad T_{yy} = \tfrac{121905}{5} - 24240.6 = 140.40, \quad T_{xy} = \tfrac{73242}{5} - 14552.4 = 96.00,$$
$$E_{xx} = 195.60, \quad E_{yy} = 206.00, \quad E_{xy} = 186.60.$$

$\hat\beta = 186.6/195.6 = 0.954$; $\hat\mu = 40.2$; $\bar x_{i\cdot} = 25.2, 26.0, 21.2$ ($\bar x_{\cdot\cdot} = 24.133$), $\bar y_{i\cdot} = 41.4, 43.2, 36.0$: $\hat\tau_1 = 1.2 - 0.954(1.067) = 0.182$, $\hat\tau_2 = 3.0 - 0.954(1.867) = 1.219$, $\hat\tau_3 = -4.2 - 0.954(-2.933) = -1.402$.

(b) $SS_E(F) = 206.0 - \tfrac{186.6^2}{195.6} = 206.0 - 178.01 = 27.99$ (11 df), $MS_E = 2.544$. Treatments: $SS_E(R) = 346.4 - \tfrac{282.6^2}{261.73} = 346.4 - 305.13 = 41.27$ (13 df); $F_0 = \tfrac{(41.27-27.99)/2}{2.544} = 2.61 < F_{0.05,2,11} = 3.98$: no machine effect after adjusting for thickness. Slope: $F_0 = \tfrac{178.01}{2.544} = 69.97 > 4.84$: thickness strongly related to strength. One-way ANOVA ignoring x: $F_0 = \tfrac{140.4/2}{206.0/12} = \tfrac{70.2}{17.17} = 4.09 > F_{0.05,2,12} = 3.89$ — it would (wrongly) attribute to the machines a difference that is explained by machine 3 producing thinner fibers ($\bar x_{3\cdot} = 21.2$).

(c) $y_\ell = \mu + \tau_1 z_{1\ell} + \tau_2 z_{2\ell} + \beta_1(x_{1\ell} - \bar x) + \beta_2(x_{2\ell} - \bar x) + \beta_3(x_{3\ell} - \bar x) + \varepsilon_\ell$, with $\mathbf{X}$ rows $(1, 1, 0, x - \bar x, 0, 0)$ for M_1, $(1, 0, 1, 0, x - \bar x, 0)$ for M_2, $(1, -1, -1, 0, 0, x - \bar x)$ for M_3; $\boldsymbol\beta = (\mu, \tau_1, \tau_2, \beta_1, \beta_2, \beta_3)^T$. Within-machine sums: M_1: $E_{xx} = 74.8$, $E_{yy} = 93.2$, $E_{xy} = 82.6$; M_2: $56.0, 54.8, 48.0$; M_3: $64.8, 58.0, 56.0$. Separate-slope SSE:

$$\left(93.2 - \tfrac{82.6^2}{74.8}\right) + \left(54.8 - \tfrac{48^2}{56}\right) + \left(58.0 - \tfrac{56^2}{64.8}\right) = 2.00 + 13.66 + 9.60 = 25.25 \ \ (9 \text{ df}).$$

$H_0 : \beta_1 = \beta_2 = \beta_3$: $F_0 = \tfrac{(27.99-25.25)/2}{25.25/9} = \tfrac{1.37}{2.81} = 0.49 < F_{0.05,2,9} = 4.26$: parallel slopes are acceptable (slopes $1.10, 0.86, 0.86$).

(d) Adjusted means $\hat\mu + \hat\tau_i$: 40.38, 41.42, 38.80. $se(\hat\tau_2 - \hat\tau_3) = \sqrt{2.544\left[\tfrac{2}{5} + \tfrac{(26.0-21.2)^2}{195.6}\right]} = \sqrt{2.544(0.4 + 0.1178)} = \sqrt{1.317} = 1.148$; difference 2.62, $t = 2.28$ ($t_{0.025,11} = 2.201$). Individually borderline, but since the overall F is not significant and a multiplicity adjustment (e.g. Bonferroni with $t_{0.05/6,11} = 2.82$) would not reject, we conclude no reliable machine differences.